



Computing the Bayesian evidence

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Plagiarism declaration

This piece of work is a result of my own work and I have complied with the Department's guidance on multiple submission and on the use of AI tools. Material from the work of others not involved in the project has been acknowledged, quotations and paraphrases suitably indicated, and all uses of AI tools have been declared.

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Location	Explanation of AI use	AI used
Code throughout the report	I used AI in order to assist me in debugging R code that computed the evidence, in particular for some of the new models in Chapters 4 and 5. However, AI did not generate the models, ideas nor approaches themselves that the code implemented.	Gemini
Figures in Chapter 3	I used AI to help me alter the colour schemes of these plots, but there was no assistance otherwise in their production.	Gemini

Abstract

This dissertation explores the evidence, a critical quantity within Bayesian statistics which is widely used for model comparison, and a number of different ways in which it can be reliably computed for a general class of models. We conduct a literature review on a number of current available methods for computing, perform a systematic analysis to assess which of the presented methods will be most accurate and computationally convenient on more realistic data, before introducing the concept of zero-inflation for count data, and applying the described models to a real-world example, looking to compute the evidence in order to compare them and determine which of the models is most likely to have generated the data. We also critically assess the notion of model comparison in such a way, and see that the idea of a closed world in which only a few models could generate a given dataset is convenient, but fails to account for the real possibility that the true model has not even been considered.

Acknowledgements

It is not exactly a secret that my final year here at Durham has been a challenging one, and it is perhaps not surprising that I have played an (admittedly) large part in contributing to that. Taking on the role of Coordinator for Nightline, one of the most intense student roles in Durham, alone, whilst balancing writing this very report and of course all else that goes on in life has pushed me to new limits, but it has allowed me to meet some of the most empathetic and understanding human beings I have the privilege to call my friends, and it has also taught me many new things about myself, for which I am incredibly grateful. Pushing through it all has not been an easy process, but I would like to thank my supervisor, Darren, for all of the help he has given me in reaching this point.

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Chapter 1

Introduction

Suppose that we have some data, which throughout this dissertation we will denote $\mathbf{y}$, which we assume to have been generated from one of a collection of models $\mathcal{M}_k$, where each model has parameters θ_k , and $k = 1, \dots, m$.

Recall that for any particular model $\mathcal{M}_k$, by Bayes' Theorem, we have that

$$\pi(\theta_k|\mathbf{y}, \mathcal{M}_k) = \frac{\pi(\theta_k|\mathcal{M}_k)\pi(\mathbf{y}|\theta_k, \mathcal{M}_k)}{\pi(\mathbf{y}|\mathcal{M}_k)},$$

and we note that the denominator here, $\pi(\mathbf{y}|\mathcal{M}_k)$, can be written as

$$\pi(\mathbf{y}|\mathcal{M}_k) = \frac{\pi(\theta_k|\mathcal{M}_k)\pi(\mathbf{y}|\theta_k, \mathcal{M}_k)}{\pi(\theta_k|\mathbf{y}, \mathcal{M}_k)}.$$

Now, we multiply both sides by the posterior and integrate over θ_k :

$$\begin{aligned} \pi(\mathbf{y}|\mathcal{M}_k)\pi(\theta_k|\mathbf{y}, \mathcal{M}_k) &= \pi(\theta_k|\mathcal{M}_k)\pi(\mathbf{y}|\theta_k, \mathcal{M}_k) \\ \Rightarrow \pi(\mathbf{y}|\mathcal{M}_k) \int_{\theta_k} \pi(\theta_k|\mathbf{y}, \mathcal{M}_k) d\theta_k &= \int_{\theta_k} \pi(\theta_k|\mathcal{M}_k)\pi(\mathbf{y}|\theta_k, \mathcal{M}_k) d\theta_k \\ \Rightarrow \pi(\mathbf{y}|\mathcal{M}_k) &= \int_{\theta_k} \pi(\theta_k|\mathcal{M}_k)\pi(\mathbf{y}|\theta_k, \mathcal{M}_k) d\theta_k, \end{aligned} \tag{1.1}$$

since $\pi(\mathbf{y}|\mathcal{M}_k)$ does not depend on θ_k and the posterior simply integrates to 1. For the purposes of this dissertation, this quantity $\pi(\mathbf{y}|\mathcal{M}_k)$ will generally be referred to as the **evidence**, however it is also often referred to as the *marginal likelihood* or *model evidence* in the literature.

The evidence has a number of useful applications, in particular within model comparison. We might, for example, be interested in computing $\pi(\mathcal{M}_k|\mathbf{y})$; that is, the posterior probability for model $\mathcal{M}_k$ given the data. Or, if we are simply interested in comparing a subset of two models directly, we might wish to calculate the Bayes Factor for the two models, which is defined as

$$B_{ij} = \frac{\pi(\mathbf{y}|\mathcal{M}_i)}{\pi(\mathbf{y}|\mathcal{M}_j)},$$

clearly relying on the calculation of the evidence. There may be other ways for us to assess the performance of models, such as the Akaike Information Criterion, but the evidence provides us with a quantifiable, and comparable metric which we can use to decide which model we favour.

Unfortunately, however, the integral (1.1) is notoriously difficult to compute, and other than in some simple situations where we have a conjugate prior setup for our model likelihood, it is not possible to calculate the integral analytically [Friel and Wyse, 2012].

This dissertation will review some of the estimators which can be employed to deal with the more difficult scenarios where models lack conjugacy in the following chapter. In Chapter 3, we

will systematically assess their performance on a more simple, tractable example in order to decide on which of the methods is most practical and accurate. Finally, in Chapters 4 and 5 we will then look at the concept of zero-inflated data, and then apply our evidence calculations to a real-world system of competing models for a set of such data, to gain an understanding of how useful the evidence can be.

Chapter 2

Computing the evidence

As [Friel and Wyse \[2012\]](#) note, the evidence is incredibly difficult to compute, since the integral (1.1) is analytically intractable in situations where the prior is not conjugate; in general, the integration may be high-dimensional, and involve extremely complex and variable functions. Therefore, a number of different numerical estimators of this quantity have been developed in order to calculate the evidence, with varying degrees of applicability in real-world problems.

For the purposes of this chapter, we will focus on the simple scenario where we focus only on one model at a time and later expand the scenario to include a system of competing models.

2.1 Running examples

Before we can look to the methods themselves, we will of course want a few simple toy examples in order to help illustrate how the estimators work and verify their results. We can use a conjugate prior setup in order to this, since as we have seen that means that (1.1) will be tractable, and thus the analytic value of the evidence calculable. We will use a very small dataset in order to make our comparisons between the true value of $\pi(\mathbf{y})$ and the different estimators' output:

$$\mathbf{y} = (0.4, 0.6, 1, -0.3, -0.9, 0.77, 0.12)^\top.$$

2.1.1 A practical note

Throughout this dissertation, we will compute the evidence on the **log-scale**, as opposed to on the natural scale. One kind of calculation that we might need to perform repeatedly throughout is to sum, for example, likelihoods on the log-scale:

$$\log \sum_{i=1}^n \pi(\mathbf{y}|\theta_i)$$

Doing this on the natural scale, however, presents a serious risk of numerical underflow, since these values may be extremely small, and risk being rounded down to zero exactly. One clever trick to do this is the Log-Sum-Exp trick, which helps to prevent underflow by working on the log-scale, and exponentiating only for the calculation of the summation itself [[Blanchard et al., 2021](#)]. We can execute this trick easily in R:

```
logSumExp <- function(logx){  
  c <- max(logx)  
  return(c + log(sum(exp(logx - c))))  
}
```

A variant on this which will simplify some of our later calculations is the Log-Mean-Exp trick, which can be implemented using the simple function below:

2.1 Running examples

```
logMeanExp <- function(logx){
  c <- max(logx)
  return(c + log(mean(exp(logx - c))))
}
```

These tricks prevent underflow, but also will simplify the computation of some of the estimators we will see later, which are defined on the log-scale. Now, we can look at the running examples used throughout this chapter.

2.1.2 Normal model with unknown mean and known variance

Consider the setup below, where we have data $y_1, \dots, y_n$, assumed to be generated independently and identically distributed (i.i.d.) from

$$\begin{aligned} Y_i | \mu, \sigma^2 &\sim \mathcal{N}(\mu, \sigma^2) \\ \mu &\sim \mathcal{N}(0, 1), \end{aligned} \tag{2.1}$$

and where the value of $\sigma^2 = 2^2$ is fixed and known.

Since we have a conjugate prior setup, we can calculate a tractable expression for the evidence of this model:

$$\begin{aligned} \pi(\mathbf{y}) &= \int_{-\infty}^{\infty} \pi(\mathbf{y} | \mu) \pi(\mu) d\mu \\ &= \int_{-\infty}^{\infty} \left(\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (y_i - \mu)^2 \right\} \right) \cdot \frac{1}{\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \mu^2 \right\} d\mu \\ &= \int_{-\infty}^{\infty} c \exp \left\{ -\frac{1}{2} \left(\frac{n\mu^2 - 2n\bar{\mathbf{y}}\mu + \sum y_i^2}{\sigma^2} + \mu^2 \right) \right\} d\mu \\ &= \int_{-\infty}^{\infty} c \exp \left\{ -\frac{1}{2} \left(A\mu^2 - 2B\mu + \frac{\sum y_i^2}{\sigma^2} \right) \right\} d\mu \\ &= \int_{-\infty}^{\infty} c \exp \left\{ \frac{B^2}{2A} - \frac{\sum y_i^2}{2\sigma^2} \right\} \cdot \exp \left\{ -\frac{A}{2} \left(\mu - \frac{B}{A} \right)^2 \right\} d\mu \\ &= c \sqrt{\frac{2\pi}{A}} \left(\frac{B^2}{2A} - \frac{\sum y_i^2}{2\sigma^2} \right), \end{aligned}$$

where throughout we have defined $A = \frac{n}{\sigma^2} + 1$, $B = \frac{n\bar{\mathbf{y}}}{\sigma^2}$, and $c = \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \cdot \frac{1}{\sqrt{2\pi}}$, and we have identified a Gaussian kernel at the last stage. Then, on the log scale, we get an expression for the evidence of

$$\log \pi_{true}(\mathbf{y}) = -\frac{n}{2} \log(8\pi) - \frac{1}{2} \log \left(\frac{n+4}{4} \right) + \frac{\left(\frac{n\bar{\mathbf{y}}}{4} \right)^2}{\frac{n+4}{2}} - \frac{\sum y_i^2}{8}.$$

With the data $\mathbf{y}$, we see that

$$\log \pi_{true}(\mathbf{y}) = -12.1364.$$

2.1.3 Normal model with unknown mean and variance

Consider a very similar case to before, but here where our variance of our data generating process σ^2 is also unknown. Such a situation will require slightly more complicated priors;

consider then again that our data $y_1, \dots, y_n$ is generated i.i.d. from the model

$$\begin{aligned} Y_i | \mu, \sigma^2 &\sim \mathcal{N}(\mu, \sigma^2) \\ \mu | \sigma^2 &\sim \mathcal{N}\left(0, \frac{\sigma^2}{\kappa}\right) \\ \sigma^2 &\sim \text{IG}(a, b), \end{aligned} \tag{2.2}$$

where here $\text{IG}(\cdot, \cdot)$ denotes the Inverse Gamma distribution, and the values of a and b can, in general, vary. For the purposes of this example, we fix $a = b = 1$. The parameter κ within the prior for the mean controls how wide our beliefs about μ are; with a lower value close to zero, the variance will be high and the likelihood will dominate the prior within the posterior distribution, and for values close to or above 1, the prior will be narrow, and will have a much greater effect on our posterior. In this example, we will set $\kappa = 0.001$.

Again, we can find a tractable expression for the evidence of this model, which, on the log scale, is

$$\begin{aligned} \log \pi(\mathbf{y}) = & -\frac{n}{2} \log(2\pi) + \frac{1}{2} \log(\kappa) - \frac{1}{2} \log(n + \kappa) + a \log(b) \\ & - \log \Gamma(a) + \log \Gamma\left(\frac{n}{2} + a\right) - \left(\frac{n}{2} + a\right) \log\left(\frac{\sum y_i^2}{2} - \frac{(n\bar{\mathbf{y}})^2}{2(n + \kappa)} + b\right). \end{aligned}$$

With the data $\mathbf{y}$, we see that

$$\log \pi_{\text{true}}(\mathbf{y}) = -12.1727.$$

Now we have two running examples to which we can compare estimates, we can move to review some of the literature in this area, and introduce a number of approaches for calculating the evidence.

2.2 The naïve approaches

There are two different, very natural approaches to attempting to estimate the evidence, which, as we shall see, are unfortunately infamously unreliable.

2.2.1 The Monte Carlo estimator

Recall that in the one-model scenario, we can write our evidence integral as

$$\pi(\mathbf{y}) = \int_{\theta} \pi(\theta) \pi(\mathbf{y}|\theta) d\theta.$$

Now using a classic Monte Carlo technique, we can view this integral as an expectation with respect to the prior distribution of θ :

$$\pi(\mathbf{y}) = \mathbb{E}_{\theta}(\pi(\mathbf{y}|\theta)).$$

As we might therefore expect, given a sample $\theta_1, \dots, \theta_n$ from the prior distribution of θ , we can calculate an estimate for the evidence based on an average of the likelihoods under each θ_i :

$$\hat{\pi}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n \pi(\mathbf{y}|\theta_i).$$

Whilst this estimate is conceptually simple, it is unfortunately famously unstable, as the likelihood required for this calculation will typically be very concentrated relative to the prior [Pajor, 2017].

2.2.2 The harmonic mean estimator

Instead then, we might wish to try and use the posterior distribution $\theta|\mathbf{y}$ to estimate our evidence instead, and this is often straightforward given modern MCMC techniques. Now, we see that

$$\begin{aligned}\mathbb{E}_{\theta|\mathbf{y}}\left(\frac{1}{\pi(\mathbf{y}|\theta)}\right) &= \int_{\theta} \frac{\pi(\theta|\mathbf{y})}{\pi(\mathbf{y}|\theta)} d\theta \\ &= \int_{\theta} \frac{\pi(\mathbf{y}|\theta)\pi(\theta)}{\pi(\mathbf{y}|\theta)\pi(\mathbf{y})} d\theta \\ &= \int_{\theta} \frac{\pi(\theta)}{\pi(\mathbf{y})} d\theta \\ &= \frac{1}{\pi(\mathbf{y})},\end{aligned}$$

since $\int_{\theta} \pi(\theta) d\theta = 1$. Then doing a little bit of rearranging, and again viewing our expectation as an average, we have the estimator

$$\hat{\pi}(\mathbf{y}) = \frac{1}{\frac{1}{n} \sum_{i=1}^n \frac{1}{\pi(\mathbf{y}|\theta_i)}},$$

where our sample $\theta_1, \dots, \theta_n$ is now sampled from the posterior $\theta|\mathbf{y}$. The main issue here is that the posterior distribution used here is determined largely by the likelihood, with the effect of the prior being relatively small. However, $\pi(\mathbf{y})$ is known to be very sensitive to changes in the prior [Neal, 1994], and thus these two properties do not align, and we have an unstable estimator as a result.

2.2.3 Running examples

Normal model with unknown mean and known variance

Looking at the model in (2.1) and considering the simple Monte Carlo estimator, we can use the code in the code repository (available in the [Appendices](#)) to calculate the evidence. In this case, we use 10,000 samples from the prior, and we see that our result is

$$\log \hat{\pi}_{MC}(\mathbf{y}) = -12.1363.$$

It is straightforward to create a Stan model to replicate this model, and we see that, using 10,000 of the resulting samples from the posterior in our harmonic mean estimator, we get the estimate

$$\log \hat{\pi}_{HM}(\mathbf{y}) = -12.1571.$$

This reassures us that our above two methods do indeed work in a simple situation, since both of the estimates returned are very close to the true, analytical result.

Normal model with unknown mean and variance

Now looking at the model defined by (2.2), we can use our two different methods to estimate the evidence with the code in the code repository with 10,000 samples, we find that our estimates

are:

$$\begin{aligned}\log \hat{\pi}_{MC}(\mathbf{y}) &= -12.2186 \\ \log \hat{\pi}_{HM}(\mathbf{y}) &= -9.5915.\end{aligned}$$

We see that our Monte Carlo estimator looks reasonably accurate, whilst the harmonic mean estimate is significantly greater than the true value — perhaps it has fallen foul of the instability previously discussed.

We have seen now that these two methods do work for these simple models, but of course we wish to expand to far more complicated models, which — as we shall see — these methods simply will not be able to handle reliably. Thus, we will need some more robust techniques.

2.3 Power posteriors

One particularly interesting method, first proposed by [Friel and Pettitt \[2008\]](#), looks at a modified version of our posterior, called the power posterior. We firstly define

$$\pi_t(\theta|\mathbf{y}) \propto \pi(\mathbf{y}|\theta)^t \pi(\theta),$$

introducing a parameter $t \in [0, 1]$ known as the **temperature parameter**. It is worth considering what role our temperature parameter plays here. When $t = 0$, this simply collapses back down to the prior distribution, and when $t = 1$, we recover our full posterior distribution, up to proportionality. So t plays a bridging role between our prior and posterior distribution, and this will be very useful for us.

Now, integrating our definition of the power posterior over θ , we can then define

$$z(\mathbf{y}|t) = \int_{\theta} \pi(\mathbf{y}|\theta)^t \pi(\theta) \, d\theta.$$

Looking at this integral, as we previously mentioned, when $t = 0$, we get our prior distribution inside the integrand, and thus our constructed value is simply 1. When $t = 1$ — assuming that $z(\mathbf{y}|t)$ is finite — we will recover the exact definition of the evidence. Now following ideas from [Gelman and Meng \[1998\]](#), and we see that on the log scale,

$$\begin{aligned}\frac{d}{dt} \log z(\mathbf{y}|t) &= \frac{1}{z(\mathbf{y}|t)} \frac{d}{dt} z(\mathbf{y}|t) \\ &= \frac{1}{z(\mathbf{y}|t)} \int_{\theta} \frac{d}{dt} \pi(\mathbf{y}|\theta)^t \pi(\theta) \, d\theta \\ &= \frac{1}{z(\mathbf{y}|t)} \int_{\theta} \pi(\mathbf{y}|\theta)^t \log \pi(\mathbf{y}|\theta) \pi(\theta) \, d\theta \\ &= \int_{\theta} \frac{\pi(\mathbf{y}|\theta)^t \pi(\theta)}{z(\mathbf{y}|t)} \log \pi(\mathbf{y}|\theta) \, d\theta.\end{aligned}$$

Again using Monte Carlo ideas previously discussed, we can then write this as an expectation:

$$\frac{d}{dt} \log z(\mathbf{y}|t) = \mathbb{E}_{\theta|\mathbf{y},t}(\log \pi(\mathbf{y}|\theta)).$$

Now we can integrate both sides of this equation over the interval $[0, 1]$, and by the Fundamental

Theorem of Calculus, we obtain

$$\begin{aligned} \int_0^1 \frac{d}{dt} \log z(\mathbf{y}|t) dt &= \log z(\mathbf{y}|t=1) - \log z(\mathbf{y}|t=0) \\ &= \log \left\{ \frac{z(\mathbf{y}|t=1)}{z(\mathbf{y}|t=0)} \right\} \\ &= \log \pi(\mathbf{y}), \end{aligned}$$

which implies that calculating $\int_0^1 \mathbb{E}_{\theta|\mathbf{y},t}(\log \pi(\mathbf{y}|\theta)) dt$ is equivalent to calculating the evidence. Unfortunately, this integral will also typically be intractable, and thus we will be required to use an approximation. In practice, the most common approach to obtaining an estimate via the power posterior method is to discretise the the integral $\int_0^1 \mathbb{E}_{\theta|\mathbf{y},t}(\log \pi(\mathbf{y}|\theta)) dt$ by choosing a particular temperature scale, $0 = t_1 < t_2 < \dots < t_m = 1$, then running an MCMC sampler for each value $t_j \in [0, 1]$, and estimating the log-evidence via a trapezoidal approximation:

$$\log \hat{\pi}(\mathbf{y}) \approx \sum_{j=0}^{m-1} (t_{j+1} - t_j) \frac{\mathbb{E}_{\theta|\mathbf{y},t_{j+1}}(\log \pi(\mathbf{y}|\theta)) + \mathbb{E}_{\theta|\mathbf{y},t_j}(\log \pi(\mathbf{y}|\theta))}{2}.$$

The choice of temperature scale is one that should not be dismissed: it can heavily impact the final result, and the accuracy of the estimator. The temperature parameter t , as discussed, forms a bridge between the prior and posterior distributions, and consequently, as t increases, the distribution will learn much more information from the likelihood for earlier, smaller values of t than larger ones, when new information will not impact the posterior distribution very much at all.

Therefore, it is important to use a temperature scale that is denser closer to the prior to enable this, and as such choose more values of t closer to 0. One way that Friel and Pettitt suggested doing this was to use a temperature scale of the form

$$t_j = \left(\frac{j}{m} \right)^c,$$

for some constant value $c > 1$, which will ensure that the scale is indeed denser closer to the prior.

This method, of course, has the main computational benefit of being entirely calculated on the log-scale, which helps to prevent numerical underflow.

2.3.1 Running examples

Normal model with unknown mean and known variance

It is not difficult to implement a setup like this within Stan, and then continuously sample from the power posterior using each temperature. For instance, in this example, we can simply alter our likelihood sampling statement such that the log-likelihood is manually updated, and multiply by the temperature parameter t of interest each time [Eriksson, 2017]:

```
// Likelihood with power posterior
target += t * normal_lpdf(y | mu, sigma);
```

As was suggested by Friel and Pettitt, we will use a geometric ladder with 50 temperatures with a value of $c = 5$, which will increase the density close to the prior as is needed for this

method to be reliable, and we will use 10,000 samples for each temperature, not including any samples discarded for burn-in. When we implement this using the code in the code repository for the data $\mathbf{y}$, we find that

$$\log \hat{\pi}_{PP}(\mathbf{y}) = -12.1379,$$

which is an error of 0.0015 with $\pi_{true}(\mathbf{y})$, suggesting good performance on this simple example.

Normal model with unknown mean and variance

We can adapt our Stan model for the unknown variance model in a very similar way, and then our log-scale estimate is returned as

$$\log \hat{\pi}_{PP}(\mathbf{y}) = -12.2124,$$

which is an error of 0.0033 with the analytic value, which is again very accurate.

2.4 Chib's method

In his now-eponymous approach to estimating the evidence, Chib [1995] rearranges the famous Bayes' Theorem statement to give, in the one-model scenario,

$$\pi(\mathbf{y}) = \frac{\pi(\mathbf{y}|\theta)\pi(\theta)}{\pi(\theta|\mathbf{y})}.$$

Now taking the logarithm of both sides, we have the estimator

$$\log \hat{\pi}(\mathbf{y}) = \log \pi(\mathbf{y}|\theta^*) + \log \pi(\theta^*) - \log \hat{\pi}(\theta^*|\mathbf{y}), \quad (2.3)$$

where, importantly, θ^* is a point of high posterior probability. Whilst the identity above holds for any θ , the estimate is more likely to be accurate and computationally stable for high posterior density θ^* than in areas of lower probability, since there will be a far greater number of θ samples in regions of high posterior density than in the tails.

In particular, θ^* is often taken to be the posterior mode or the maximum likelihood estimator (MLE) to ensure that it is indeed of high posterior probability. Whilst it is possible to use the posterior mean, we must be careful to ensure that it is indeed of high probability; in a multimodal distribution, for instance an additive mixture model, the mean will not necessarily be a point of high probability, and thus in general it is often easier to rely on the mode or MLE.

We can see this through a simple example: Consider two toy normal models:

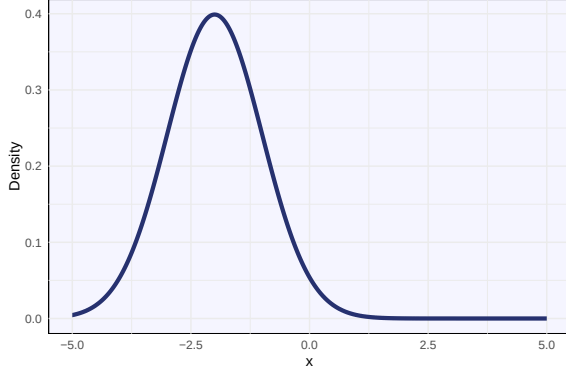
$$Y_1 \sim \mathcal{N}(-2, 1)$$

$$Y_2 \sim \mathcal{N}(2, 0.5^2)$$

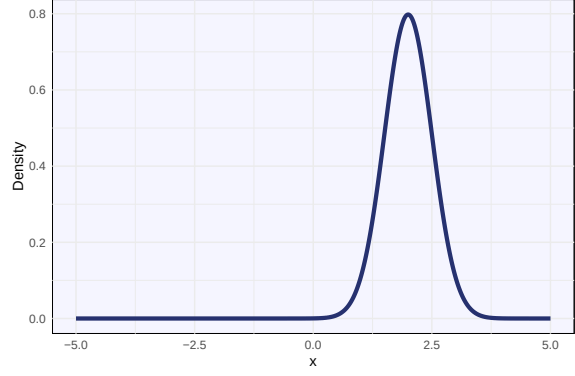
Figures 2.1a and 2.1b show these distributions. We can then easily form an equally weighted additive mixture model, displayed in Figure 2.1c. Here we see that the resulting distribution is, of course, multimodal. Whilst it is not difficult to tell in this particular instance given the setup that the mode of the mixture model is 2, the mean will no longer align with this mode, as it would usually do in the case of a singular normal distribution, since it is pulled leftwards by the first spike in density from the distribution of Y_2 .

We notice from identity (2.3) that the quantities $\log \pi(\mathbf{y}|\theta^*)$ and $\log \pi(\theta^*)$ can be easily computed as the densities required will be available for sampling from. Chib's main contribution was to realise that one can use the output of a Gibbs sampler to estimate the final quantity, $\log \hat{\pi}(\theta^*|\mathbf{y})$.

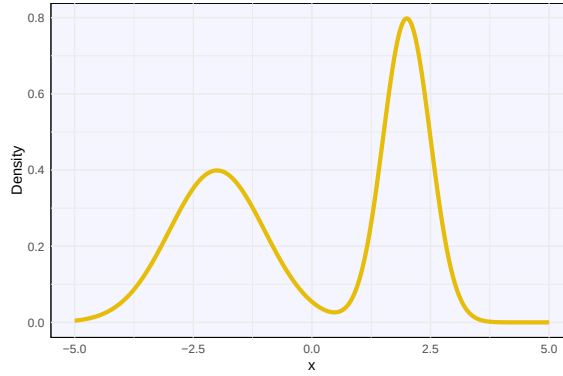
2.4 Chib's method



(a) Plot of the distribution of Y_1



(b) Plot of the distribution of Y_2



(c) Plot of the resulting additive mixture model

2.4.1 Running the Gibbs Sampler

Suppose we have a distribution with parameter θ , which can be partitioned as $\theta = (\theta_1, \theta_2)^\top$. Then we can write

$$\pi(\theta|\mathbf{y}) = \pi(\theta_1|\theta_2, \mathbf{y})\pi(\theta_2|\mathbf{y}).$$

We note here that $\pi(\theta_1|\theta_2, \mathbf{y})$ is a full conditional density (FCD), and so we can use a Gibbs sampler to evaluate the density, however the latter density is not an FCD, because it does not depend on θ_1 . However, we can construct this FCD by using an average over the θ_1 samples. By using the law of total probability, we can write

$$\pi(\theta_2|\mathbf{y}) = \int \pi(\theta_2|\theta_1, \mathbf{y})\pi(\theta_1|\mathbf{y}) d\theta_1.$$

Now, we can view this as a Monte Carlo expectation and write

$$\pi(\theta_2|\mathbf{y}) = \mathbb{E}_{\theta_1|\mathbf{y}}(\pi(\theta_2|\theta_1, \mathbf{y})),$$

and then approximating this expectation using the output of a Gibbs sampler taking draws $\theta_1 \sim \pi(\theta_1|\mathbf{y})$ we have the approximation

$$\pi(\theta_2|\mathbf{y}) \approx \frac{1}{G} \sum_{g=1}^G \pi(\theta_2|\theta_1^{(g)}, \mathbf{y}),$$

where G is the total number of samples obtained from the Gibbs sampler after discarding burn-in. Thus we have what we need, computed using the FCD $\pi(\theta_2|\theta_1, \mathbf{y})$.

We can see this here for a simpler situation with only two parameter blocks to update, but it is not difficult to see how this can be extended to a more general situation with p parameter blocks; for a parameter $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)^\top$, using the law of total probability, we can write

$$\pi(\boldsymbol{\theta}|\mathbf{y}) = \pi(\theta_1|\theta_2, \dots, \theta_p, \mathbf{y})\pi(\theta_2|\theta_3, \dots, \theta_p, \mathbf{y}) \cdots \pi(\theta_p|\mathbf{y}).$$

Then, extending the process of approximating the required densities by averaging the FCDs over the posterior samples:

$$\begin{aligned}\pi(\theta_2|\theta_3, \dots, \theta_p, \mathbf{y}) &\approx \frac{1}{G} \sum_{g=1}^G \pi(\theta_2|\theta_1^{(g)}, \theta_3, \dots, \theta_p, \mathbf{y}) \\ \pi(\theta_3|\theta_4, \dots, \theta_p, \mathbf{y}) &\approx \frac{1}{G} \sum_{g=1}^G \pi(\theta_3|\theta_1^{(g)}, \theta_2^{(g)}, \theta_4, \dots, \theta_p, \mathbf{y}) \\ &\vdots \\ \pi(\theta_p|\mathbf{y}) &\approx \frac{1}{G} \sum_{g=1}^G \pi(\theta_p|\theta_1^{(g)}, \dots, \theta_{p-1}^{(g)}, \mathbf{y}).\end{aligned}$$

It is, of course, very useful that we can obtain the required densities in this way, even for high-dimensional problems, and indeed the Gibbs sampler can be applied to a very wide variety of models, which makes this method incredibly powerful.

2.4.2 Running examples

It makes little sense here to consider the basic scenario of only one unknown parameter, as the Gibbs sampler will be trivial.

Normal model with unknown mean and variance

Using the setup as before, we see that we will need the following components in order to calculate the identity (2.3):

- $\pi(\boldsymbol{\theta}) = \pi(\mu, \sigma^2) = \pi(\mu|\sigma^2)\pi(\sigma^2)$
- $\pi(\mathbf{y}|\boldsymbol{\theta}) = \pi(\mathbf{y}|\mu, \sigma^2)$
- $\pi(\boldsymbol{\theta}|\mathbf{y}) = \pi(\mu|\sigma^2, \mathbf{y})\pi(\sigma^2|\mathbf{y})$

Here, we can see that $\pi(\mathbf{y}|\mu, \sigma^2)$ is just the likelihood of the data, and then using the conjugacy of our priors, we have that the required FCDs for this calculation are

$$\begin{aligned}\mu|\sigma^2, \mathbf{y} &\sim \mathcal{N}\left(\frac{n\bar{\mathbf{y}}}{n+\kappa}, \frac{\sigma^2}{n+\kappa}\right) \\ \sigma^2|\mu, \mathbf{y} &\sim \text{IG}\left(\frac{a+n}{2}, \frac{b+\sum_{i=1}^n (y_i - \mu)^2 + \kappa\mu^2}{2}\right)\end{aligned}$$

Then, as we described above, we will need to approximate the density $\pi(\sigma^2|\mathbf{y})$ by averaging over the μ samples using the Gibbs sampler output:

$$\pi(\sigma^2|\mathbf{y}) \approx \frac{1}{G} \sum_{g=1}^G \pi(\sigma^2|\mu^{(g)}, \mathbf{y}).$$

2.5 Summary

Now, we can use the code in the code repository to evaluate all of these densities at a point $(\mu^*, \sigma^{2*})^\top$ of high posterior probability; in particular, we can create our Gibbs sampler using the following function, taking advantage of the `invgamma` package [Kahle and Stamey, 2016]:

```
gibbs_complex <- function(N, data, a, b, kappa = 0.001, pmean = 0) {  
  mat <- matrix(ncol = 2, nrow = N)  
  datamean <- mean(data)  
  n <- length(data)  
  mu <- 0  
  sigma_sq <- 1  
  mat[1, ] <- c(mu, sigma_sq)  
  for (i in 2:N)  
  {  
    mu <- rnorm(1, (n*datamean)/(n + kappa), sqrt(sigma_sq/(n + kappa)))  
    sigma_sq <- rinvgamma(1, a + n/2 + 0.5, b + (sum((data - mu)^2)  
      + (kappa*(mu - pmean)^2))/2)  
    mat[i, ] <- c(mu, sigma_sq)  
  }  
  return(mat)  
}
```

Doing this with 10,000 samples yields a result of

$$\log \hat{\pi}_{Chib}(\mathbf{y}) = -12.1729,$$

which is the closest estimate we have seen, suggesting that this may be a highly reliable approach to use.

2.5 Summary

In this chapter, we have seen that due to the intractability of integral (1.1), we will need to turn to different methods in order to compute the evidence, and have thus seen a number of different methods for performing this estimation. Whilst the Monte Carlo and harmonic mean methods are perhaps easiest to understand intuitively, they are not always the most reliable predictors. Using power posteriors and Chib's method may well be more reliable, but exactly how much more reliable are they, and how well do these estimators perform when we vary the number of samples they can work with?

Chapter 3

A systematic analysis

We have now seen a number of different methods for estimating the evidence, but it would of course be useful to know which one is the most useful and practical for more complicated models, as well as what the limits of these methods are.

One way in which we will explore the behaviours of each of these different estimators would be to vary the number of samples that are drawn (from either the prior or posterior distributions, depending on the estimator) and examine the variance and bias displayed. Before we can perform this systematic analysis, however, we will of course need a more complicated model to examine, and for this we can use a Bayesian linear regression model setup.

3.1 Setting the scene

3.1.1 The penguins data

For our analysis, we will use data on various attributes of adult penguins on three islands in the Palmer Archipelago, Antarctica. The measurements include various aspects of their size (flipper length, body mass, bill dimensions), and sex for Adelie, Chinstrap, and Gentoo penguins [Gorman et al., 2014]. In this analysis, we will focus on the effect of bill length (`bill_len`), bill depth (`bill_dep`) and flipper length (`flipper_len`) on the penguins' body mass (`body_mass`).

Since the data is on a very large scale, we will first scale this data to have zero mean and unit variance, which will make inference significantly easier. We can have a quick first look at the scaled data by producing a pairs plot and investigating the effects on the response. To do this whilst accounting for potential differences between species, we will use the `GGally` package [Schloerke et al., 2025] to produce a more advanced `ggpairs` plot, which is given in Figure 3.1.

We can see that the variables here do indeed appear to be linearly affecting the penguins' body mass, with a strongly positive association, but also some clustering affecting the data, particularly with the effect of bill depth on the mass, which is indeed caused by the differences in species. Given these strong linear associations, we see that a linear model is worthy of investigation, and so we can fit a Bayesian linear regression model to this data.

3.1.2 The Bayesian linear regression model

The simple Bayesian linear regression is, in general, determined by the following model setup [Gundersen, 2020]:

$$\begin{aligned} Y|\mathbf{X}, \boldsymbol{\beta}, \sigma^2 &\sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}) \\ \boldsymbol{\beta}|\sigma^2 &\sim \mathcal{N}(\boldsymbol{\mu}_0, \sigma^2 \boldsymbol{\Lambda}_0^{-1}) \\ \sigma^2 &\sim \text{IG}(a_0, b_0) \end{aligned}$$

3.1 Setting the scene

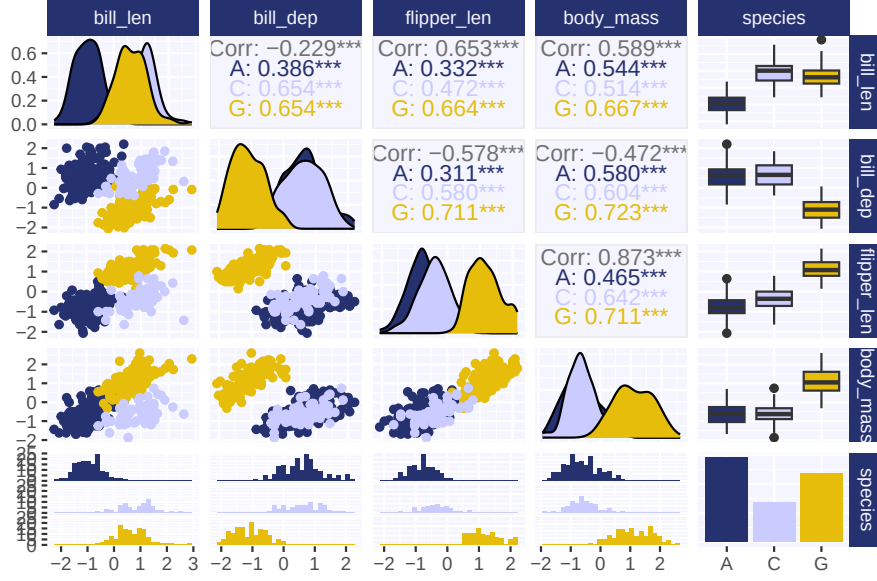


Figure 3.1: `ggpairs` plot of the scaled penguins data.

where

- $\mathbf{X}$ is the design matrix, which includes the data from all covariates and a column with $\mathbf{X}_{i1} = 1$ to account for an intercept within the model,
- β is the vector of covariate coefficients,
- μ_0 is the prior mean of β ,
- Λ_0^{-1} is the prior precision associated with β , and
- σ^2 is the observation error.

The response is, in general, determined by a normal distribution whose mean is the ‘standard’ linear predictor, along with observational error. The Bayesian core of the model is that the vector of predictor coefficients β is also given a prior normal distribution, and the observational error σ^2 is given an inverse-gamma prior. The resulting effect is that this model is conjugate, which allows us not only to easily calculate the posterior distribution, but also in this case to analytically calculate the evidence.

The posterior analytic evidence

Given the model parameters above, O’Hagan [1994] showed that, given data $y_1, \dots, y_n$ and $\mathbf{X}$, the form of the posterior distribution for this model is

$$\beta, \sigma^2 | \mathbf{y}, \mathbf{X} \propto \pi(\beta | \mathbf{y}, \mathbf{X}, \sigma^2) \pi(\sigma^2 | \mathbf{y}, \mathbf{X}),$$

where we have that

$$\begin{aligned} \beta | \mathbf{y}, \mathbf{X}, \sigma^2 &\sim \mathcal{N}(\mu_n, \sigma^2 \Lambda_n^{-1}) \\ \sigma^2 | \mathbf{y}, \mathbf{X} &\sim \text{IG}(a_n, b_n). \end{aligned}$$

The posterior parameters for this model setup are then

$$\begin{aligned}\mathbf{\Lambda}_n &= \mathbf{X}^\top \mathbf{X} + \mathbf{\Lambda}_0, \\ \boldsymbol{\mu}_n &= \mathbf{\Lambda}_n^{-1} (\mathbf{\Lambda}_0 \boldsymbol{\mu}_0 + \mathbf{X}^\top \mathbf{y}), \\ a_n &= a_0 + \frac{n}{2}, \\ b_n &= b_0 + \frac{1}{2} (\mathbf{y}^\top \mathbf{y} + \boldsymbol{\mu}_0^\top \mathbf{\Lambda}_0 \boldsymbol{\mu}_0 - \boldsymbol{\mu}_n^\top \mathbf{\Lambda}_n \boldsymbol{\mu}_n).\end{aligned}$$

Turning to the evidence calculation, we then have that

$$\pi(\mathbf{y}) = (2\pi)^{-\frac{n}{2}} \sqrt{\frac{|\mathbf{\Lambda}_0|}{|\mathbf{\Lambda}_n|}} \cdot \frac{b_0^{a_0}}{b_n^{a_n}} \cdot \frac{\Gamma(a_n)}{\Gamma(a_0)},$$

or on the log-scale,

$$\begin{aligned}\log \pi(\mathbf{y}) &= -\frac{n}{2} \log(2\pi) + \frac{1}{2} \log |\mathbf{\Lambda}_0| - \frac{1}{2} \log |\mathbf{\Lambda}_n| \\ &\quad + \log b_0^{a_0} - \log b_n^{a_n} + \log \Gamma(a_n) - \log \Gamma(a_0).\end{aligned}\quad (3.1)$$

Application to the penguins data

For the penguins data, we are interested in modelling the scaled body masses of each of the $n = 333$ penguins using their bills' lengths and depths, as well as their flipper lengths. We will use relatively noninformative priors in this case, setting

$$\begin{aligned}a_0 &= 2, \\ b_0 &= 1, \\ \boldsymbol{\mu}_0 &= \mathbf{0}, \\ \mathbf{\Lambda}_0^{-1} &= \begin{pmatrix} \frac{1}{10} & 0 & 0 & 0 \\ 0 & \frac{1}{10} & 0 & 0 \\ 0 & 0 & \frac{1}{10} & 0 \\ 0 & 0 & 0 & \frac{1}{10} \end{pmatrix},\end{aligned}$$

taking into account the scaling of the data and being conservative about any interactions between covariates at this stage.

Computing the analytical value of the log-evidence for this model, we obtain

$$\log \pi_{true}(\mathbf{y}) = -250.493. \quad (3.2)$$

Immediately we can see that even on the log-scale this is an incredibly small number, and so with regard to the different methods we have seen for computing an estimate of the evidence, we will need to be careful to avoid numerical underflow, or our calculations will be perturbed significantly. Thus, we will make significant use of the Log-Sum-Exp and Log-Mean-Exp tricks.

3.2 Comparing the estimators

3.2.1 Designing the analysis

Now that we have calculated the analytical value of the log-evidence for this model on this dataset, we are interested in comparing the performance of the different estimators that were

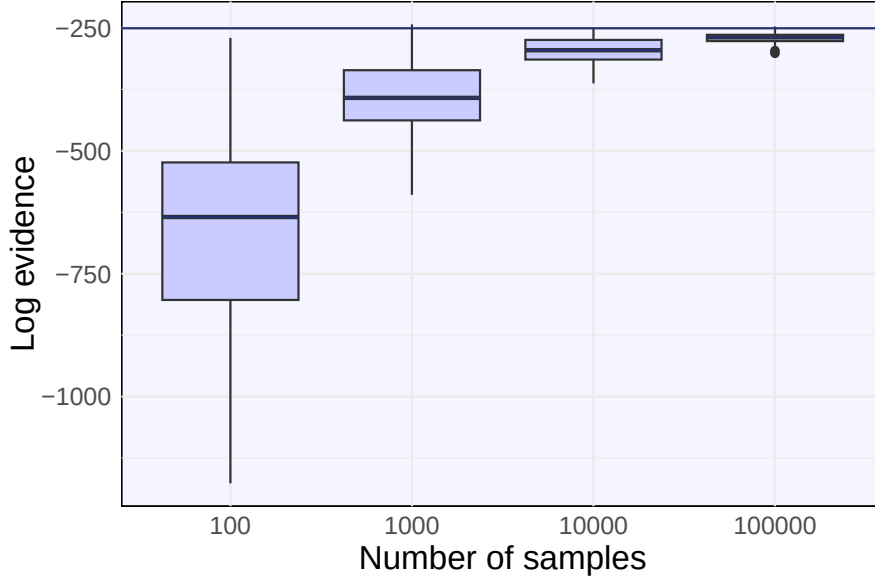


Figure 3.2: Boxplot of the Monte Carlo estimates, varying the number of samples.

introduced in Chapter 2, and comparing them to the true value. This is only possible here because of the conjugate setup of the Bayesian linear regression model, but it will be useful in informing us of which of the different estimators may be best to use for problems and models whereby the analytic value of the log-evidence is intractable, as it is for the vast majority of cases.

To do this, we can vary the number of samples which are drawn from the relevant distribution (either the prior or posterior distribution, depending on the estimator) and assess the estimates generated. Doing this many times will then give us a distribution of estimates for each number of samples drawn, which we can then use to more effectively and systematically assess the efficacy of the estimators, comparing them with the true value in (3.2).

For each of the estimators, we will compare the results of 10^i samples, where $i = 2, 3, 4, 5$ in turn, and we will repeat each value of i 100 times to generate a useful distribution of estimates.

3.2.2 The Monte Carlo estimator

Recall that the Monte Carlo estimator of the evidence is, in general,

$$\hat{\pi}_{MC}(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n \pi(\mathbf{y}|\theta_i).$$

In this context, and given the parameters of this model, we have that on the log-scale,

$$\log \hat{\pi}_{MC}(\mathbf{y}) = -\log n + \log \sum_{i=1}^n \pi(\mathbf{y}|\beta_i, \sigma_i^2).$$

When we perform our analysis on the Monte Carlo estimator for this setup, we obtain the distribution of estimates shown in Figure 3.2. The analytic value obtained in (3.2) is shown on the boxplot in blue. We can see immediately that the number of samples provided is crucial; when only 100 samples are used, the estimator is far from the true analytic value, and it has undershot significantly. As more samples are provided, we can see that the estimator begins

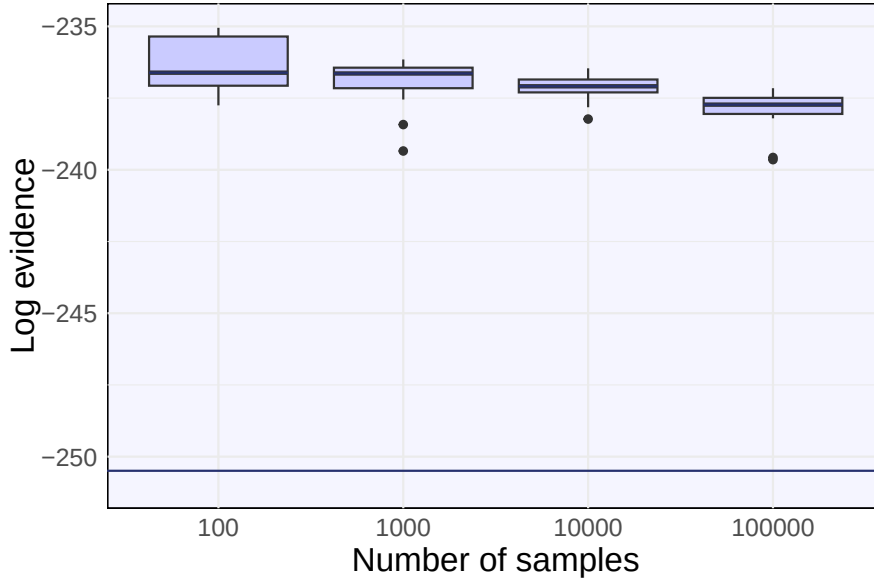


Figure 3.3: Boxplot of the harmonic mean estimates, varying the number of samples.

to converge on the true value, although it is only reaching a somewhat reasonable estimate after around 100,000 samples. Indeed, the median of the estimates when 100,000 samples were provided was $\log \hat{\pi}_{MC}(\mathbf{y}) = -267.34$, which is an error of around 6.3% with the true analytic value. Computationally, this is achievable for this relatively simplistic model, but for models with a higher number of parameters this could become more taxing.

Also worthy of note is the behaviour of the variance of the estimator: as the number of samples increases, the variance decreases as it closes in on the right value. This is reassuring, as we would expect any reasonable estimator to become more confident as it is provided with more and more data.

Overall, the Monte Carlo estimator has not been as accurate as we would like despite its computational ease on this example. Indeed, its extension into more complex models may be even less reliable or convenient, given how poorly it performed for this model with few samples.

3.2.3 The harmonic mean estimator

In the context of the Bayesian linear regression model setup, the log-scale harmonic mean estimator is

$$\log \hat{\pi}_{HM}(\mathbf{y}) = \log n - \log \sum_{i=1}^n \frac{1}{\pi(\mathbf{y}|\boldsymbol{\beta}_i, \sigma_i^2)}$$

In order to use Stan to generate MCMC samples from the posterior distribution here, we will need to define the multivariate normal distribution for $\boldsymbol{\beta}$ in terms of the precision matrix $\mathbf{A}_0$. To do this, we will use the `multi_normal_prec` distribution statement in Stan [Stan Development Team, 2026a] to prevent the need for repeated matrix inversion, which can be computationally expensive, particularly for large matrices.

Running the analysis for the harmonic mean estimator, we obtain the boxplot shown in Figure 3.3. For this estimator, we can see two problems very quickly:

- **Inaccuracy:** Immediately we can see that the estimator is overshooting significantly; indeed, even after 100,000 samples are provided to it, the median of the log-evidence is $\log \hat{\pi}_{HM}(\mathbf{y}) = -237.73$, which is an error of about 5.1% with the analytic value. This is

of course not as accurate as we would wish.

- **Variability:** What is perhaps more worrying is the lack of stability here; as the sample size increases, the variance does not monotonically decrease, suggesting that the estimator is not confident in its predictions.

The instability particularly is a direct result of the misalignment discussed in Section 2.2.2; Neal [1994] noted that because the posterior is determined significantly by the likelihood, but the evidence is sensitive to changes in the prior, we get the instability that we can now visualise in the boxplot.

Overall then, we can conclude that it is not worth considering the harmonic mean estimator for simpler problems like these, let alone a more complicated one. Its instability means it cannot be relied upon for accurate results, and so we will seek to use other methods for more realistic scenarios.

3.2.4 Power posteriors

Recall from Chapter 2 that the power posterior is a modified version of the posterior, with a temperature parameter $t \in [0, 1]$ placed on the likelihood component:

$$\pi_t(\theta|\mathbf{y}) \propto \pi(\mathbf{y}|\theta)^t \pi(\theta).$$

In this context, we have that the log-scale power posterior estimate of the evidence is

$$\int_0^1 \mathbb{E}_{\beta, \sigma^2|\mathbf{y}, \mathbf{X}, t}(\log \pi(\mathbf{y}|\mathbf{X}, \beta, \sigma^2)) dt;$$

in practice, however, this integral is intractable, and so instead we can discretise the interval over which t runs. In this case, we will use the discretisation suggested by Friel and Pettitt [2008]; that is, the geometric temperature ladder

$$t_j = \left(\frac{j}{m}\right)^c,$$

where $c > 1$ is a constant. Here, we will choose $m = 25$ and $c = 5$, such that the ladder is denser closer to the prior. Thus the computation being done to calculate each estimate is

$$\log \hat{\pi}_{PP}(\mathbf{y}) \approx \sum_{j=0}^{m-1} (t_{j+1} - t_j) \frac{\mathbb{E}_{\beta, \sigma^2|\mathbf{y}, \mathbf{X}, t_{j+1}}(\log \pi(\mathbf{y}|\mathbf{X}, \beta, \sigma^2)) + \mathbb{E}_{\beta, \sigma^2|\mathbf{y}, \mathbf{X}, t_j}(\log \pi(\mathbf{y}|\mathbf{X}, \beta, \sigma^2))}{2}.$$

Again, here we will need to draw upon the `multi_norm_prec` distribution statement within Stan in order to compute the required MCMC samples.

The distribution of the estimates, varying for the number of samples, is shown in Figure 3.4. We can see that the distribution of estimators covers the analytic value of the evidence even with just 100 samples, though it is worth noting that these MCMC chains are themselves unstable given their small size. As the number of samples grows, we can see that it converges on a particular value, and gets increasingly more confident in its prediction, as we would like. One trade-off, however, is that of compute time and power: we are generating a large number of fresh MCMC samples for each iteration, which is taxing in terms of compute time. In particular, for more complex models than this, this could become a severe bottleneck on standard hardware, even if we are interested in only computing a single estimate of the evidence, rather than conducting the analysis seen here.

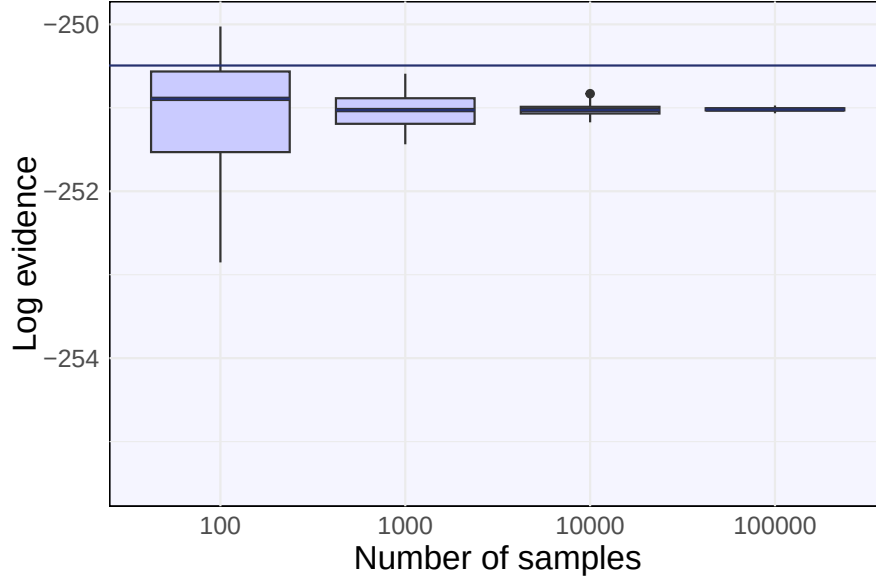


Figure 3.4: Boxplot of the power posterior estimates, varying the number of samples.

It is noticeable, though, that the value on which the estimator has converged is below the true value of the log-evidence. This bias has been introduced because the value of m , the number of points on our discretised temperature ladder, has been kept constant throughout, and not increased as the number of samples is increased. We can take a look at this effect in more detail.

Turning up the heat

In order to attempt to fix this bias issue with the power posterior method, we can increase the number of temperatures on the ladder as we increase the number of samples. To do this, instead of fixing $m = 25$, we will choose a sequence $m_j = 10j, j = 1, 2, 3, 4$. When we now implement the varying temperature ladder, we get a boxplot that is shown in Figure 3.5.

We can see that the estimates begin much further away from the analytic value, though this is due to a reduced number of temperatures for these smaller sample sizes. As hoped, the estimates for the higher sample sizes have now increased in accuracy as the temperature ladder has increased in size, and the estimator's confidence has also grown, with the variability of the predictions shrinking towards zero quickly as the number of samples increases. We could further increase the value of m to obtain an even more accurate estimator, however this comes with the trade-off that more compute time will be required.

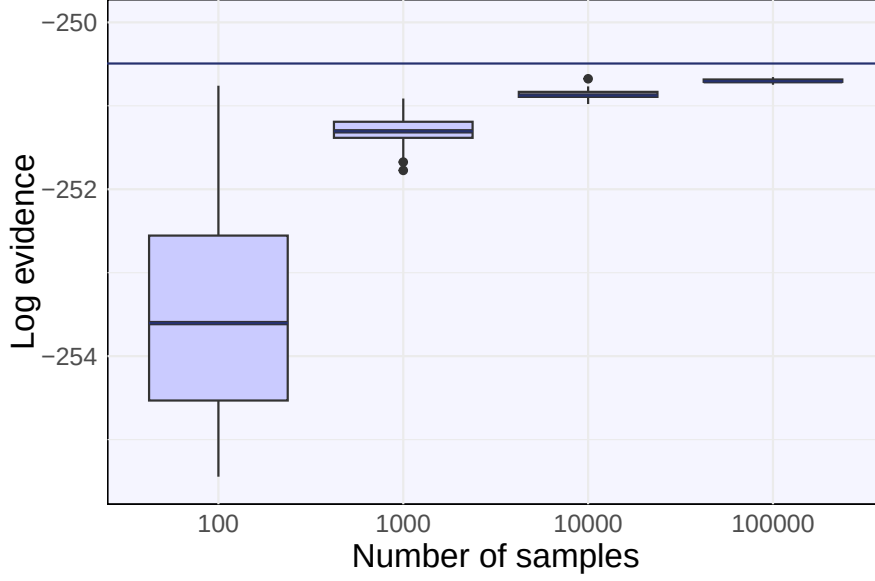


Figure 3.5: Boxplot of the updated power posterior estimates, varying the number of samples and the temperature ladder.

3.2.5 Chib's method

Chib's identity for the calculation of the log-evidence is, for this model setup,

$$\begin{aligned} \log \hat{\pi}_{Chib}(\mathbf{y}) &= \log \pi(\mathbf{y}|\theta^*) + \log \pi(\theta^*) - \log \hat{\pi}(\theta^*|\mathbf{y}) \\ &= \log \pi(\mathbf{y}|\beta^*, \sigma^{2*}) + \log \pi(\beta^*, \sigma^{2*}) - \log \hat{\pi}(\beta^*, \sigma^{2*}|\mathbf{y}). \end{aligned}$$

Here, we will need to ensure that our parameters β^* and σ^{2*} are of high posterior density; since the normal and inverse-gamma distributions are unimodal, in this case we can use the mean of the resulting Gibbs samples. As discussed in Section 2.4, we already know the densities $\pi(\mathbf{y}|\beta^*, \sigma^{2*})$ and $\pi(\beta^*, \sigma^{2*})$, since

$$\pi(\beta^*, \sigma^{2*}) = \pi(\beta^*|\sigma^{2*})\pi(\sigma^{2*}),$$

our model's prior densities. As for the posterior, we have that

$$\pi(\beta^*, \sigma^{2*}|\mathbf{y}) = \pi(\beta^*|\sigma^{2*}, \mathbf{y})\pi(\sigma^{2*}|\mathbf{y}).$$

The former density is one which we already know, and the second we can estimate by averaging over the draws of a Gibbs sampler; that is,

$$\hat{\pi}(\sigma^{2*}|\mathbf{y}) = \frac{1}{G} \sum_{g=1}^G \pi(\sigma^{2*}|\beta^{(g)}, \mathbf{y}).$$

Now, we further have that

$$\pi(\sigma^2|\beta, \mathbf{y}) \propto \pi(\mathbf{y}|\beta, \sigma^2)\pi(\beta|\sigma^2)\pi(\sigma^2),$$

and so we can use this to find the parameters for the inverse-gamma full conditional density. Suppose that we are at state $1 \leq g \leq G$ of the update chain. Focussing on the updated shape parameter $a^{(g)}$ first, we see that the terms which will contribute are:

- $(\sigma^2)^{-(a_0+1)}$ from the prior for σ^2 ,
- $(\sigma^2)^{-\frac{n}{2}}$ from the likelihood, and
- $(\sigma^2)^{-\frac{p}{2}}$ from the prior for β ,

where p is the number of model coefficients. Overall, the update for the first parameter is thus

$$a^{(g)} = a_0 + \frac{n}{2} + \frac{p}{2}.$$

Similarly for our remaining parameter $b^{(g)}$, we have that the terms which contribute are those terms within the exponentials, which will update the parameter by using the most recent $\beta^{(g)}$:

- $\exp\left\{-\frac{b_0}{\sigma^2}\right\}$ from the prior for σ^2 ,
- $\exp\left\{-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta^{(g)})^\top (\mathbf{y} - \mathbf{X}\beta^{(g)})\right\}$ from the likelihood, and
- $\exp\left\{-\frac{1}{2\sigma^2}(\beta^{(g)} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Lambda}_0 (\beta^{(g)} - \boldsymbol{\mu}_0)\right\}$ from the prior for β .

Overall, the update for the second parameter is therefore

$$b^{(g)} = b_0 + \frac{1}{2} \left((\mathbf{y} - \mathbf{X}\beta^{(g)})^\top (\mathbf{y} - \mathbf{X}\beta^{(g)}) + (\beta^{(g)} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Lambda}_0 (\beta^{(g)} - \boldsymbol{\mu}_0) \right).$$

Applying our Gibbs sampler and performing the systematic analysis over the Chib estimates, we obtain the boxplot shown in Figure 3.6.

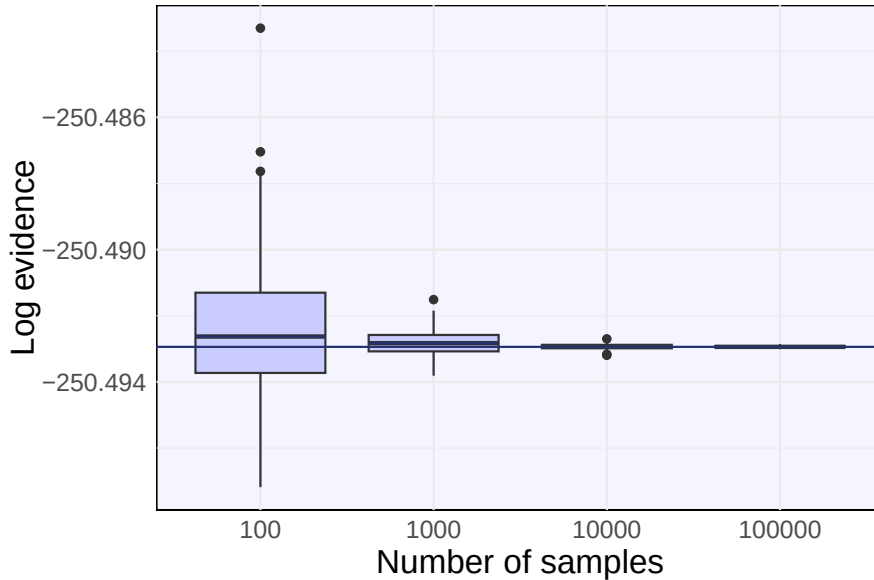


Figure 3.6: Boxplot of Chib estimates, varying the number of samples.

The first thing to notice here is the y -axis of this boxplot: this is an incredibly accurate estimator. Indeed, the percentage error between the median of the estimates with the highest sample size and the analytic log-evidence is $2.43 \times 10^{-6}\%$ — effectively negligible. Moreover, we also see that there is good convergence onto a particular value, and that the variance in the estimates decreases monotonically as we would like.

3.3 Summary

However, whilst Chib’s method here is accurate and not particularly computationally expensive (we have only run a single Gibbs sampler), it is more difficult to implement because of the need to know all of the full conditional densities. As we have discussed, this is of course possible here, but for a model with a large number of different parameters, this could become a taxing task.

3.2.6 A direct comparison

As a final effort to compare the estimators directly, we can now fix the number of samples provided to each estimator at 100,000 and compare the different distributions of the estimates more directly, and the resulting boxplot is shown in Figure 3.7. In this comparison, we have used $m = 40$ temperatures for the power posterior estimator, as in Section 3.2.4.

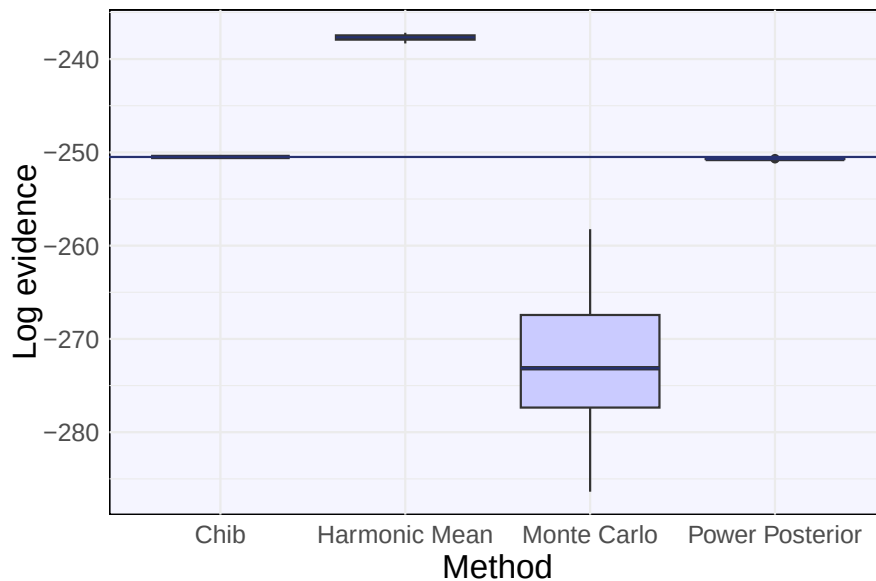


Figure 3.7: Boxplot comparing the different estimators, for a fixed sample size of 100000.

This reveals to us the obvious differences in the estimators: clearly here the most accurate ones are the Chib estimator and the power posterior estimator, which are extremely close to the analytical value. The harmonic mean, however, has significantly overshoot the true evidence, whilst the Monte Carlo estimate is not only significantly smaller than the analytic evidence, but even with this relatively high number of samples the variance of the estimates produced is still much higher than for the other estimators.

3.3 Summary

In this chapter, we have compared the different methods to compute the evidence in a more systematic way, in order to determine which estimator we should focus on when looking at more complicated models whose analytical evidence is intractable. We have seen that the more naïve methods, the Monte Carlo and harmonic mean approaches, will not be reliable to use even for the Bayesian linear regression model seen here, let alone more complicated models. The estimates produced are of high variance, and sit far away from the true value.

In comparison, for higher numbers of samples, Chib’s estimator and the power posterior method perform very well; Chib’s method in particular produces very accurate results even with few samples. There is, however, a practical trade-off between the two different ways of performing

the computation: focussing on pure accuracy would lead one to prefer Chib's estimator, as it runs quickly and without absorbing high amounts of compute time. However, particularly for non-standard model setups, whilst it will always be theoretically possible to construct the distributions required for the Gibbs sampler and final calculation, using the power posterior method may prove easier to communicate. Despite its more expensive computation, it can be implemented easily, simply by adapting an existing MCMC sampler. Thus, in our subsequent analyses, we will use the power posterior method in order to calculate the evidence.

Chapter 4

Complex models: zero-inflated counts

One major kind of data used in the real world is count data — how many items in a manufacturing process are defective, how many times in a month will a Londoner use the Underground, or how many contact hours any given Durham University student will have on a particular weekday. Whilst data like this will of course have many uses, often times count data will have a unique problem: the number of zeroes. If we ask a manufacturer of computer parts how many defective CPUs they produce on an hourly basis across a week, there will be many hours with no defects at all. Similarly, many Londoners will simply not use the Tube regularly because they do not live near a station, and some Durham students have very few contact hours across their entire term, let alone on one day.

This presents a unique problem: how do we model this kind of data? Typically, we might think to model counts using a Poisson distribution, or perhaps a binomial or negative binomial distribution, but ultimately these largely lack the ability to incorporate what [Lambert \[1992\]](#) described as a ‘perfect’ state. Take for example manufacturing defects: small changes in an environment could cause the entire underlying process to move between an imperfect state where defects may arise on the production line at random, and a perfect state where there are no defects. Such a process will of course lead to an inflated number of zero counts being observed given the lack of defects.

If we suppose that the response variable (the number of defects, in the example above) depends on a number of covariates, then we might think to use a Poisson regression setup to model this situation, but as we will see, this fails to account for the zero-inflation on its own. Moreover, where the data we observe are overdispersed (that is, the variance of the data is greater than the mean), we run into further issues with fitting, as a Poisson distributional assumption attempts to keep the mean and variance roughly equal.

Instead, in this chapter, we will examine several different ways in which we can account for zero-inflation within count data, compare their efficacy, and use the different models to compute the evidence by taking advantage of the ease of the power posterior method; we can then use our estimates to meaningfully compare the models.

4.1 Zero-inflated Poisson models

When dealing with count data with a single response $\mathbf{y}$ that we wish to model based on some covariates $\mathbf{X}$, we may immediately turn to a Poisson Generalised Linear Model (GLM). We recall [\[Nguyen, 2025\]](#) that

$$\begin{aligned} \mathbf{Y} &\sim \text{Po}(\boldsymbol{\mu}) \\ \boldsymbol{\mu} &= e^{\boldsymbol{\eta}} \\ \boldsymbol{\eta} &= \mathbf{X}\boldsymbol{\beta}, \end{aligned}$$

so that in this model, we are using the natural link (the log-link) and the linear predictor for each observation η_i is simply $\beta^\top \mathbf{x}_i$, with β being our vector of covariate coefficients. This is a useful model in general, but it does not account for any zero-inflation of the data.

4.1.1 Inflating the zeroes

Instead, Lambert suggested the Zero-Inflated Poisson (ZIP) model, whereby a proportion p of the data are pulled from the ‘perfect’ state, and are exactly zero. The data $\mathbf{Y}$ are therefore generated by

$$Y_i \sim \begin{cases} 0 & \text{with probability } p \\ \text{Po}(\mu_i | \eta_i) & \text{with probability } 1 - p. \end{cases}$$

Now, because of course the Poisson process itself can still generate zeroes, we have that the total probability of observing a zero under this process is $p + (1 - p)\mathbb{P}(Y_i = 0 | \mu_i) = p + (1 - p)e^{-\mu_i}$. Thus overall,

$$Y_i = \begin{cases} 0 & \text{with probability } p + (1 - p)e^{-\mu_i} \\ k & \text{with probability } (1 - p)\frac{e^{-\mu_i}\mu_i^k}{k!}. \end{cases}$$

for $k = 1, 2, \dots$. This new model now allows for an underlying proportion of observations p to be drawn as zero, whilst keeping a Poisson distributional assumption for the remainder of the data. Some of the zeroes will be from the ‘perfect’ state, and there was never any chance of them being anything other than zero (for example, a Londoner who does not live near the Tube!), and some will be drawn from the Poisson process (just by random chance). Here, we have defined the model such that p is a single number, and there is a certain chance of any observation being a perfect zero, but of course the chance of a perfect zero may depend on other covariates (like distance to the nearest Tube station), so we could further generalise this approach in order to compute a unique perfect-state probability p_i for each observation. For ease in further analysis, however, we will continue with the constant case.

4.1.2 A sanity check: do ZIP models actually work?

We can compare a regular Poisson GLM and a ZIP model to see the improved zero-inflation estimation in action. To do this, we will generate some synthetic data, which will allow us to ensure that estimates of the model coefficients are indeed in line with the true values used to generate the data.

We will consider a single response $\mathbf{y}$ which depends on three covariates, $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, as well as including an intercept. Before we see any data at all, we will of course need to choose prior distributions for the model parameters. Because we are generating our own synthetic data, we of course will actually be choosing the correct values here, but we suppose that we do not know them for now, and choose relatively uninformative priors. For each $i = 1, 2, 3, 4$, we have

$$\beta_i \sim \mathcal{N}(0, 1.5^2)$$

For the data, the three covariates will be drawn from a standard normal distribution, and we will draw $n = 340$ observations of the $\mathbf{x}_i$. Choosing the true model parameters

$$p_{true} = 0.3, \\ \beta_{true} = (1.2, 0.5, -0.4, 0.3)^\top,$$

4.1 Zero-inflated Poisson models

we can then generate our synthetic data from this setup: for each observation, with probability p_{true} , we draw a zero, and with probability $1 - p_{true}$, we draw from the $Po(e^{\beta^\top x_i})$ distribution. A histogram showing the distribution of the draws is shown in Figure 4.1.

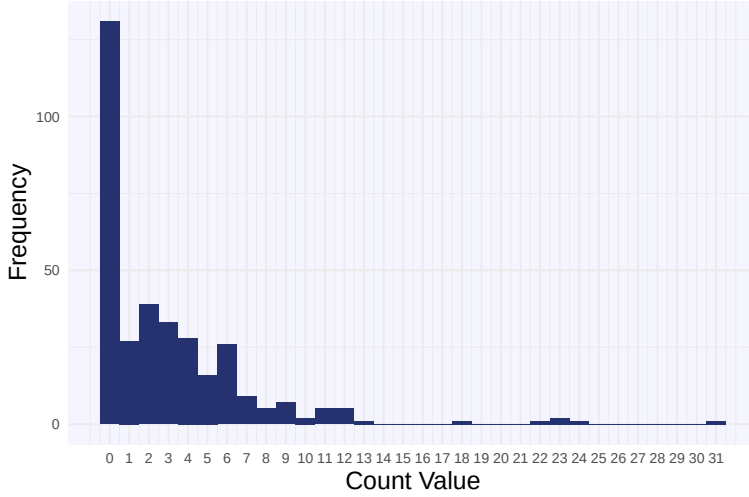


Figure 4.1: Histogram of the zero-inflated Poisson-generated data.

ulations of the data using the fitted parameters. Figure 4.2 compares the distribution of the newly simulated counts versus the counts generated earlier.

As we might expect, the predictions are indeed woeful; they fail to take into account any zero-inflation, and so non-zero values are severely over-estimated, whilst there are insufficient zero counts.

Next, we will sample from the posterior of the ZIP model. We will of course also need a prior for our perfect state proportionality parameter p , which we will choose to be $p \sim \text{Unif}[0, 1]$ — again, rather uninformative. To perform the sampling, we will write our model in Stan in the following way [Stan Development Team, 2026a]:

```
model {
  // Priors
  real p = inv_logit(logit_p);
  vector[n] log_eta = x * beta;

  beta ~ normal(0, 1.5);
  logit_p ~ logistic(0, 1);
}
```

A 30% rate of perfect zeroes leads, as we would expect, to a large spike in the data at zero — we can tell already that a standard Poisson regression model will struggle here.

To prepare for the evidence calculation that we are ultimately interested in, we will draw from the posterior distribution using MCMC; in particular, we will set the sampler up for the power posterior method, but only draw samples with $t = 1$ such that we are drawing directly from the posterior distribution. Once we have sampled from the posterior distribution using the Poisson regression likelihood, we can compute sim-

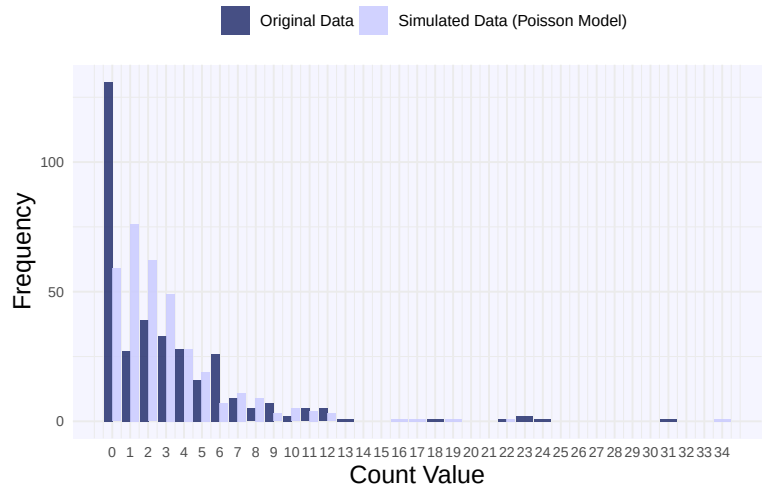


Figure 4.2: Histogram displaying the new simulated counts from the Poisson model versus the true data.

```
// Likelihood

for (i in 1:n){
  real log_lik_i;
  if (y[i] == 0){
    log_lik_i = log_sum_exp(log(p),
      log1m(p) + poisson_log_lpmf(y[i] | log_eta[i]));
  }
  else {
    log_lik_i = log1m(p) + poisson_log_lpmf(y[i] | log_eta[i]);
  }
  target += t * log_lik_i;
}
}
```

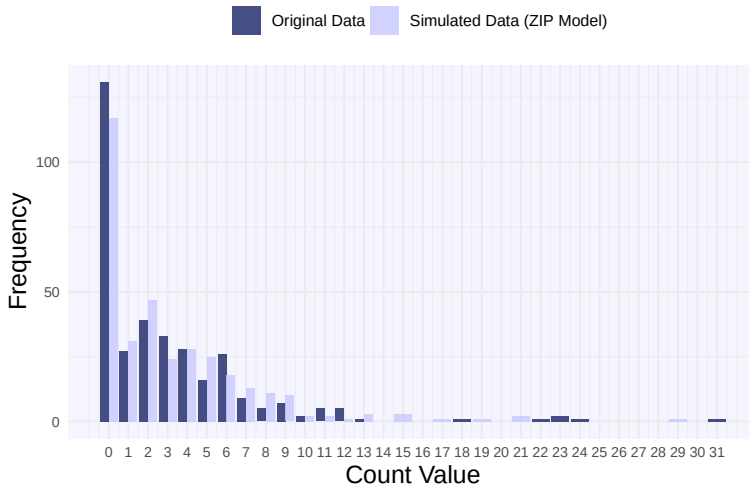


Figure 4.3: Histogram displaying the simulated counts from the ZIP model versus the true data.

Here, note that we used an equivalent distribution statement for `logit_p` as opposed to `p` — this, along with the use of `log_sum_exp()` and `log1m()` (which computes $\log(1 - p)$) is used for computational stability and to cut down on run time. Computing the simulations of the count data using the fitted values from this model, we obtain a distribution of counts shown in Figure 4.3.

This is immediately much better and more accurate to the generated data: the inflation of the zeroes has been reflected in the model, whilst the

Poisson process for the distribution of the remaining counts has been preserved.

It is reassuring to see that the ZIP model has taken account of the zero-inflation well, but is it the best model setup we can use to handle zero-inflated count data? We can also investigate another parametrisation.

4.2 Zero-inflated negative binomial models

A different way to view our Poisson GLM is by directly placing a prior on the mean of the Poisson distribution μ_i ; in fact, we can place a Gamma prior on the Poisson rate. In particular, defining ϕ to be the dispersion parameter for the data, we will use the model

$$\begin{aligned} Y_i | \mu_i &\sim \text{Po}(\mu_i) \\ \mu_i | \phi, \eta_i &\sim \text{Ga}\left(\phi, \frac{\phi}{\eta_i}\right), \end{aligned} \quad (4.1)$$

where again $\eta_i = \beta^\top \mathbf{x}_i$ is the linear predictor. The main advantage of this particular model setup is that, marginally, the y_i follow a negative binomial distribution [Gundersen, 2019]. When

4.2 Zero-inflated negative binomial models

using the negative binomial distribution for regression, as in this case, one defines the probability mass function in terms of its mean [Hilbe, 2011]. For the general NB(r, p) distribution, we have that the mean is $m = \frac{r(1-p)}{p}$, and some simple rearranging thus leads to $p = \frac{r}{m+r}$ and $1 - p = \frac{m}{m+r}$. We must also extend the domain of r such that it can take non-integer values for regression. Therefore, the binomial coefficient in the probability mass function becomes

$$\binom{k+r-1}{k} = \frac{(k+r-1)(k+r-2) \cdots (r)}{k!} = \frac{\Gamma(k+r)}{k!\Gamma(r)}.$$

We can then use this formulation to prove that the marginal distribution of the Y_i will indeed be negative binomial:

$$\begin{aligned} \pi(y_i) &= \int_0^\infty \pi(y_i|\mu_i) \pi(\mu_i|\eta_i, \phi) d\mu_i \\ &= \int_0^\infty \frac{e^{-\mu_i} \mu_i^{y_i}}{y_i!} \cdot \frac{\left(\frac{\phi}{\eta_i}\right)^\phi}{\Gamma(\phi)} \mu_i^{\phi-1} e^{-\left(\frac{\phi}{\eta_i}\right)\mu_i} d\mu_i \\ &= \frac{\left(\frac{\phi}{\eta_i}\right)^\phi}{\Gamma(\phi)y_i!} \int_0^\infty \mu_i^{\phi+y_i-1} e^{-\left(1+\frac{\phi}{\eta_i}\right)\mu_i} d\mu_i, \end{aligned}$$

and then spotting a Gamma kernel, we have

$$\begin{aligned} &= \frac{\left(\frac{\phi}{\eta_i}\right)^\phi}{\Gamma(\phi)y_i!} \cdot \frac{\Gamma(\phi+y_i)}{\left(1+\frac{\phi}{\eta_i}\right)^{\phi+y_i}} \\ &= \frac{\Gamma(\phi+y_i)}{\Gamma(\phi)y_i!} \cdot \left(\frac{\phi}{\eta_i}\right)^\phi \left(1+\frac{\phi}{\eta_i}\right)^{-(\phi+y_i)} \\ &= \frac{\Gamma(\phi+y_i)}{\Gamma(\phi)y_i!} \left(\frac{\phi}{\eta_i+\phi}\right)^\phi \left(\frac{\eta_i}{\eta_i+\phi}\right)^{y_i} \\ &= \text{NB}(\eta_i, \phi). \end{aligned}$$

Now we have established the parametrisation for our model, we can move to incorporate the zero-inflation.

4.2.1 Inflating the zeroes

Following the same idea as Lambert's ZIP model, we can incorporate the 'perfect' state using a proportionality parameter p . Using the marginal negative binomial model (4.1), we have that the Y_i will be distributed according to

$$Y_i \sim \begin{cases} 0 & \text{with probability } p \\ \text{NB}(\eta_i, \phi) & \text{with probability } 1 - p. \end{cases}$$

For this model, we have that η_i is the expected mean, and ϕ takes the role of r . Therefore, applying this for our particular distribution, we have that

$$Y_i = \begin{cases} 0 & \text{with probability } p + (1-p) \left(\frac{\phi}{\eta_i+\phi}\right)^\phi \\ k & \text{with probability } (1-p) \frac{\Gamma(k+\phi)}{k!\Gamma(\phi)} \left(\frac{\eta_i}{\eta_i+\phi}\right)^k \left(\frac{\phi}{\eta_i+\phi}\right)^\phi \end{cases}$$

for $k = 1, 2, \dots$. This model now draws the y_i marginally from a negative binomial distribution, but includes the possibility of zeroes being drawn from the perfect state.

4.2.2 A sanity check: do ZINB models actually work?

Again we will generate synthetic data using the same overall setup as in Section 4.1.2, but this time generating the response values using the zero-inflated negative binomial (ZINB) model. In choosing a true value of ϕ , we will set $\phi_{true} = 2$ in order to generate some data with a fairly high overdispersion. The histogram of the generated distribution is shown in Figure 4.4.

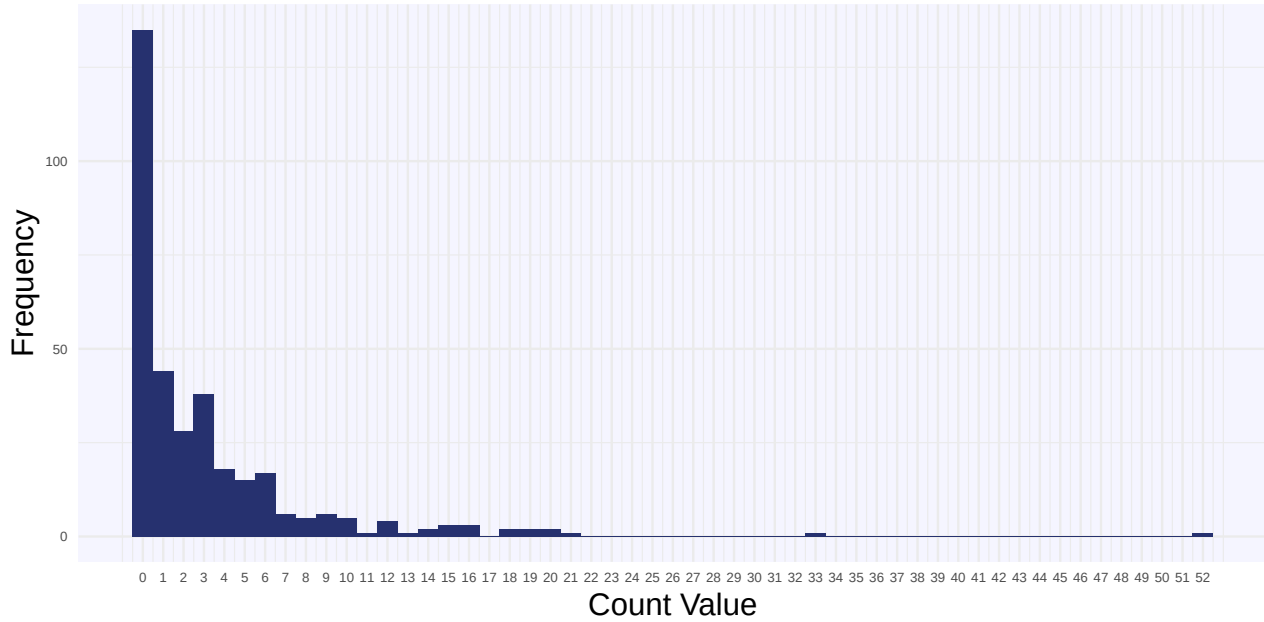


Figure 4.4: Histogram of the zero-inflated negative binomial-generated data.

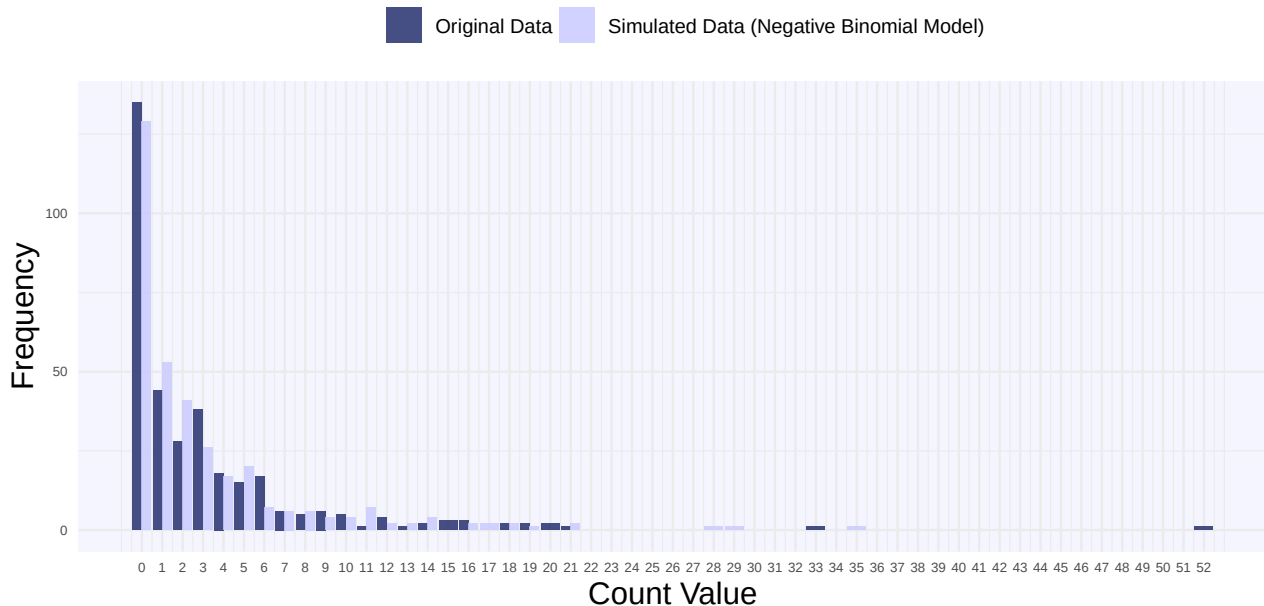


Figure 4.5: Histogram displaying the new simulated counts from the negative binomial model versus the true data.

The data is similar to previously, but with a slightly larger spike at zero. We can implement the MCMC sampler for the model (4.1) using the `neg_binomial_2_log_lpmf` distribution statement

4.2 Zero-inflated negative binomial models

for the regression parametrisation of the negative binomial distribution [Stan Development Team, 2026b]. Simulating count values using the fitted parameters, we get a distribution of counts shown in Figure 4.5.

Here we actually see that the negative binomial distribution on its own already accounts for a large amount of the zero inflation, but that it tends to slightly overestimate the frequency of the smaller non-zero counts. Generally though, this seems much better.

We can now sample from the posterior of our ZINB model, and again we will place the same priors on our perfect state parameter p and coefficients β_j as we did for the ZIP model. We will of course also require a prior for ϕ ; given that this is the dispersion of the data, we want to ensure that we are not limiting the possible posterior values of ϕ that could be obtained, and so setting the prior distribution $\phi \sim \text{Exp}(0.5)$ allows for the possibility of overdispersion in the data, whilst still giving relatively large probability to a dispersion of about 1. The Stan model code to sample from this model is given below.

```
model {  
  // Priors  
  real p = inv_logit(logit_p);  
  vector[n] log_eta = x * beta;  
  
  beta ~ normal(0, 1.5);  
  logit_p ~ logistic(0, 1);  
  phi ~ exponential(0.5);  
  
  // Likelihood  
  
  for (i in 1:n){  
    real log_lik_i;  
  
    if (y[i] == 0){  
      log_lik_i = log_sum_exp(log(p),  
        log1m(p) + neg_binomial_2_log_lpmf(y[i] | log_eta[i], phi));  
    }  
    else {  
      log_lik_i = log1m(p) + neg_binomial_2_log_lpmf(y[i] | log_eta[i], phi);  
    }  
  
    target += t * log_lik_i;  
  }  
}
```

The histogram in Figure 4.6 shows the realisations from the posterior, compared to the actual generated data. The difference between the ZINB model and the original model without the zero-inflation is actually more subtle here — again, we see good estimation of the number of zero counts, and the overestimation of the frequency of counts 1 and 2 has been reduced slightly. Whilst this difference is overall relatively small, it still seems worthy of further use.

What is particularly noteworthy in general about these two negative binomial models is how much better they perform in the situation here where the data are overdispersed significantly. The mean of the synthetic data is just 3.12, whereas the variance is 27.02, so the data most

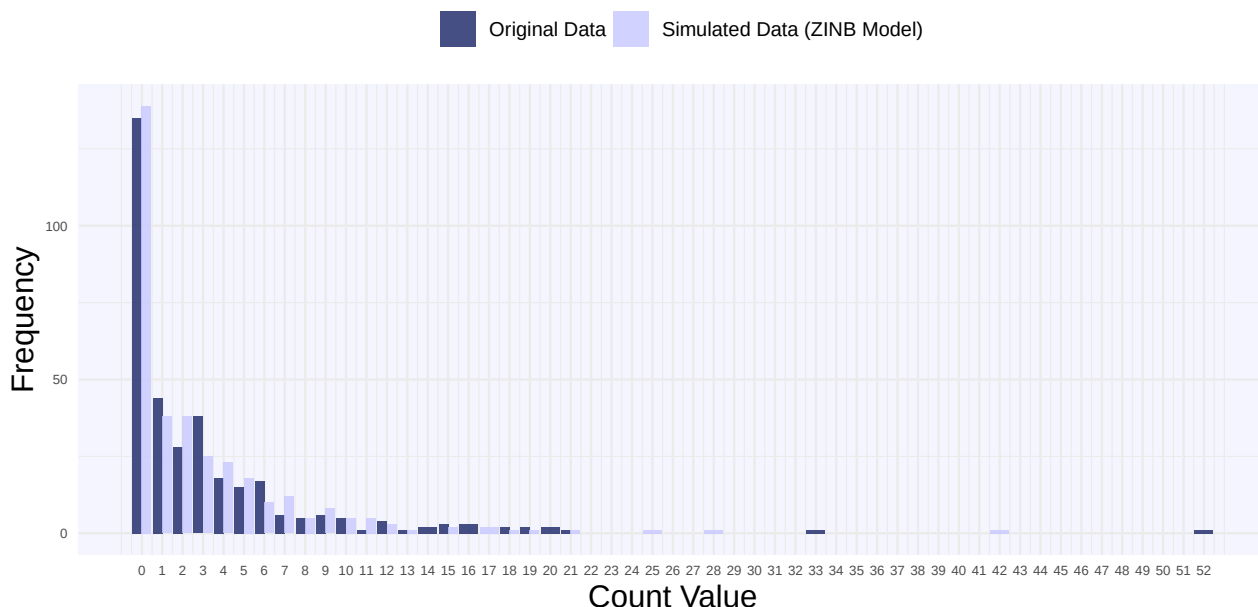


Figure 4.6: Histogram displaying the simulated counts from the ZINB model versus the true data.

certainly fall into this category, and it is visibly similar for the synthetic data generated for Section 4.1. Indeed, whilst a Poisson model is typically fitted for data that is relatively evenly dispersed, for overdispersed data such as this, it will be useful to us to consider fitting a negative binomial model; the direct inclusion of a dispersion parameter ϕ helps to account for this.

4.3 Computing the evidence

Of course, we are interested in computing the evidence for these models, and as we will see in the next chapter, these values can then be used to directly compare the competing models in a quantitatively meaningful way. The ZIP and ZINB models that we have explored in this chapter are, at their heart, mixture models; we combine the perfect state that generates only zeroes with an imperfect state that is represented by a probability distribution. This clearly leads to a model setup whereby our analytic evidence is not tractable, and this is where our estimators show their power: they can estimate the value of the evidence despite this lack of tractability. In particular, as we concluded previously, we can easily adapt the power posterior method for use with a very wide range of different model setups, including these ones.

Since we have already set up each of our models in this chapter to incorporate the temperature parameter t that is required for the power posterior method, we can very easily compute the log-evidence estimate using the same trapezoidal integral approximation as usual. We consider the synthetic data generated by the zero-inflated negative binomial in Figure 4.4, and we can apply compute the required log-evidence estimates for each of the four models, which are given in Table 4.1.

Looking at the values, we can see a significant difference in the log-evidence between, for example, the base Poisson model and the ZINB model. This is course not surprising given that the data were synthetically generated by a zero-inflated negative binomial process, and we could go further by examining these differences between all of the models; this is equivalent to computing the Bayes factor, as we will shortly see.

4.4 Summary

Model	Estimated log-evidence
Base Poisson	-1027.91
ZIP	-764.5035
Base negative binomial	-717.2376
ZINB	-702.5277

Table 4.1: Values of the log-evidence estimates for the different models.

This computation is, on the whole, not particularly taxing. We will see in the next chapter ways in which we can use the values of the evidence estimates in order to compute posterior model probabilities and Bayes factors, which will allow us to compare these models directly.

4.4 Summary

In this chapter, we have seen a number of different possible ways to handle a complicated real-world issue: that of zero-inflated count data. Whilst we might intuitively think that a Poisson regression model would be the best way to deal with this, we have seen that it is necessary to adapt the model to more effectively account for the zero-inflation by introducing a mixture component which outputs zero with some perfect state probability, which performs relatively well. Perhaps even more useful is the Gamma-Poisson parametrisation of the model; the counts become marginally negative binomially distributed, and we saw that when we account for the zero-inflation in a similar fashion, data that was simulated from the model's posterior was very promising in how close it appeared to the original data, and how the value of p adapted given the synthetic data.

In the next chapter, we will look to fit competing models to a real-life zero-inflated count dataset, and compare them by using the evidence to compute posterior model probabilities and Bayes factors.

Chapter 5

Applying the evidence

Now that we have seen how we can handle zero-inflated data, which models work best for it, and how we can compute the evidence, we can move to real-world application. In particular, given a set of zero-inflated data, how can we directly compare the a set of competing models in a Bayesian way? This chapter will focus on applying the methods that we have seen to real-world data.

5.1 Setting the scene

For our analysis in this chapter, we will look at selection of data from the 2022 U.S. National Household Travel Survey [[U.S. Department of Transportation, 2023](#)], and in particular our response variable will be the number of times survey respondents ordered groceries via grocery delivery services in the 30 days prior to their response. The full dataset available contains 16,997 responses and 153 variables; for the purposes of our analysis, we will investigate the response `DELIV_GROC`, as well as the following potentially influential variables:

- `R_AGE`: Respondent age (in years)
- `R_SEX`: Respondent sex — a factor, either “Male” or “Female”
- `EDUC`: Respondent’s highest level of education — a factor with 8 levels, ranging from “Less than high school” to “Professional/Doctorate degree”
- `WORKER`: Respondent’s employment status — a factor, either “Worker” or “Not worker”
- `MEDCOND`: Whether or not the responder has a condition or disability that makes travel difficult — a factor, either “Yes” or “No”
- `DRIVER`: The responder’s driver status — a factor, either “Yes” or “No”

We note that, since most of these variables are factors, we will have a total of 13 parameters to investigate the significance of. A full breakdown of the variable encoding is given in the [Appendices](#).

Whilst there were nearly 17,000 responses, many participants were not asked all questions due to differing eligibility criteria for each question. To ensure that our analysis is fair across all of the different variables, we will only take into account those responses where data is available for all six covariates and the response; this gives us a sample size of $n = 5553$. The response `DELIV_GROC` is of course count data, and we can have an initial look at the data in [Figure 5.1](#). We see that the data is indeed heavily zero-inflated; given the cost of grocery delivery services and their lack of availability everywhere in the United States, this is perhaps not surprising. Moreover, the data are overdispersed, with a mean count of 0.76, but a variance of

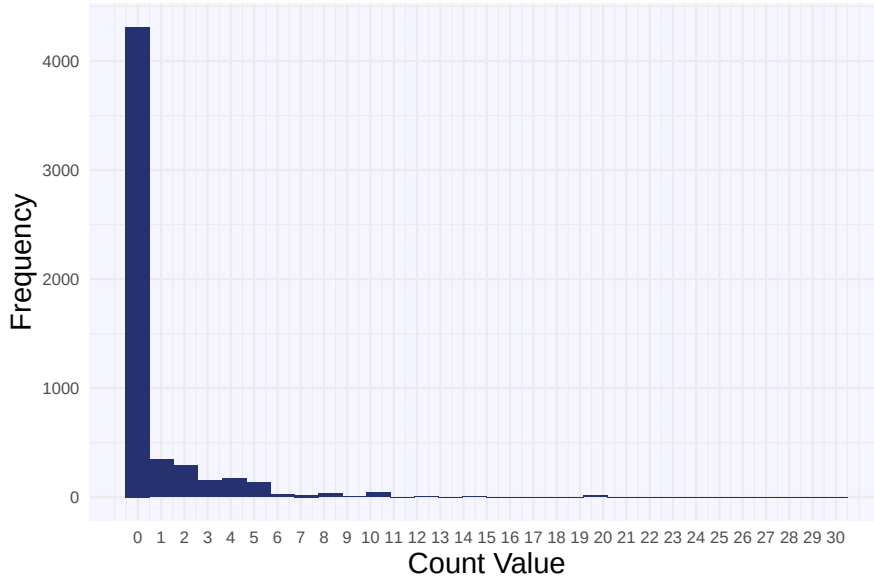


Figure 5.1: Number of times respondents used grocery delivery services within the previous 30 days.

4.14. Therefore, we will need to turn to the models discussed in the previous chapter in order to be able to model this data effectively.

Thus, we will suppose that we have a system of 4 competing models, $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$, to which we will assign the Poisson regression, ZIP regression, negative binomial regression and ZINB regression models from before respectively. We can then fit them on this real data, and, using a log link, regress on the six covariates above.

5.2 Fitting the models

Before we look to direct model comparison, we can look to visualise the posterior distributions for each of the competing models, in order to visually assess their performance.

5.2.1 Poisson models

We will firstly fit our two Poisson models, one including the zero-inflation and one without. We will fit the models using a power posterior MCMC sampler, and again simply set the temperature parameter to $t = 1$ such that we sample from the posterior. We will use 10,000 samples (once burn-in has been discarded) to estimate the posterior model parameters.

Before we can fit the models though, we will of course need to place priors on our model coefficients β and (for the ZIP model) the perfect state

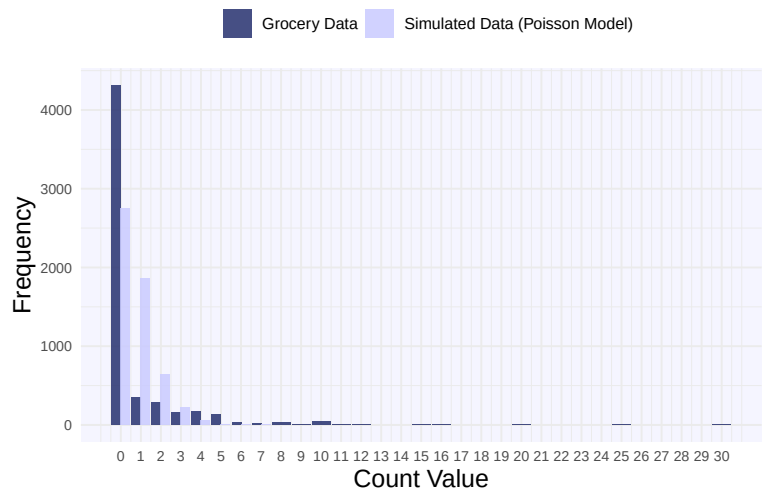


Figure 5.2: Comparison of the grocery data with the values simulated using the Poisson model.

probability p . Since we know relatively little about the effects that each covariate will have on the response, we will set the same non-informative normal prior for each of the coefficients, and a non-informative logistic prior for $\text{logit}(p)$, as we did in Section 4.1. That is, for $j = 1, \dots, 13$,

$$\beta_j \sim \mathcal{N}(0, 1.5^2)$$

$$\text{logit}(p) \sim \text{Logistic}(0, 1).$$

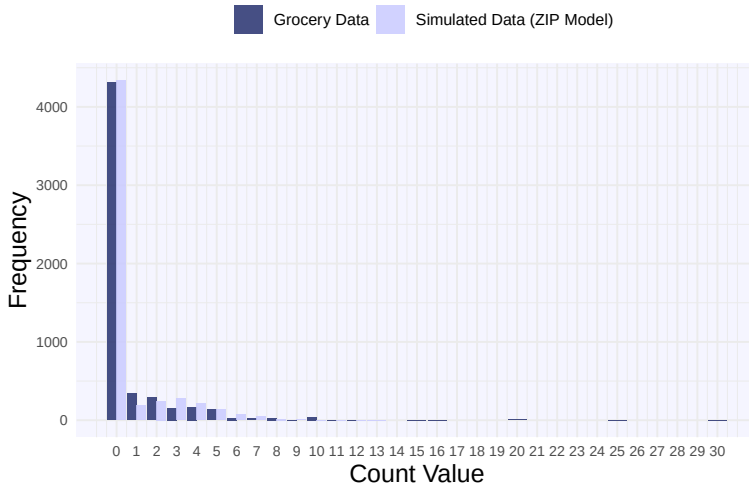


Figure 5.3: Comparison of the grocery data with the values simulated using the ZIP model.

Similarly to as we did in the previous chapter, once we have fit the two models on this data, we can simulate data from the posterior distributions in order to visually assess how well the models have performed at accounting for the zero-inflation. Figure 5.2 shows the simulations using the posterior parameters from the Poisson model compared to the actual data, and 5.3 shows the simulations using the posterior ZIP model parameters.

As we expect, the ZIP model accounts for the zero-inflation well, and the posterior value of the perfect state probability is $p = 0.766$, which suggests a very high proportion of respondents whose responses are drawn from a zero state which the Poisson distribution alone cannot account for. Indeed, the base Poisson model once again produces woeful simulations in comparison to the real data. We will therefore expect in our model comparison using the evidence later that the ZIP model is preferred over the first model which does not account for any zero-inflation.

5.2.2 Negative binomial models

We will now move to our negative binomial models, again fitting one model which accounts directly for the zero-inflation, and another model which does not. We saw previously that the negative binomial distribution alone allows the model to account for more zero-inflation, so it is of interest to us to assess just how much adding in the zero-inflation to the model is actually useful. We will of course be able to determine this later with our evidence calculations.

Recalling the model setup (4.1), we will need to place a prior on the dispersion parameter ϕ . We will make use of the same relatively non-informative prior $\phi \sim \text{Exp}(0.5)$. After we have sampled from

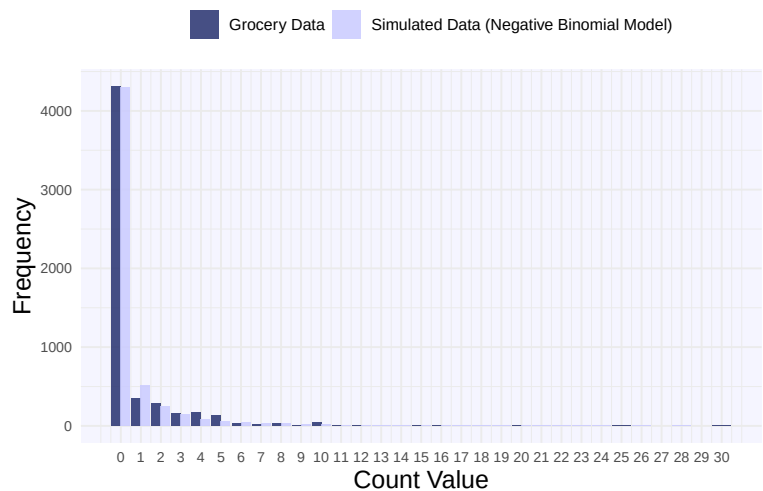


Figure 5.4: Comparison of the grocery data with the values simulated using the negative binomial model.

5.3 Model comparison: computing the evidence

the posterior distributions, we can again visualise the simulations using the posterior model parameters, and compare the simulations to the true data in order to initially assess the models' performance. Figures 5.4 and 5.5 show the simulated data based on the posterior model parameters from the negative binomial and ZINB models respectively.

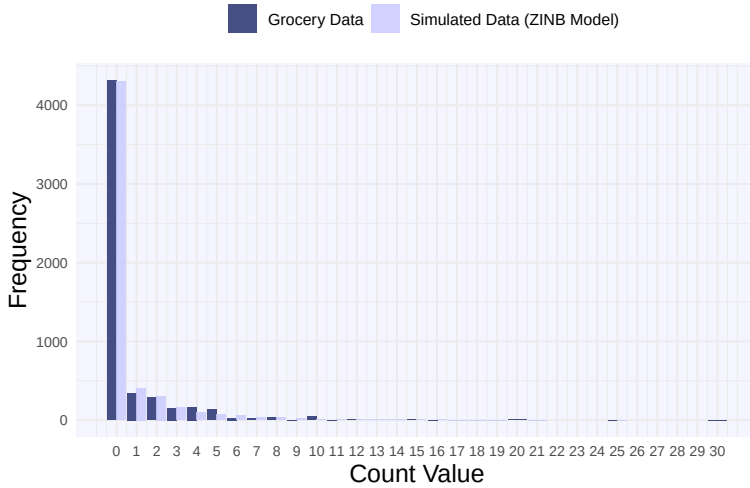


Figure 5.5: Comparison of the grocery data with the values simulated using the ZINB model.

As we expected, there is not a significant difference we can determine visually between the two negative binomial models, with perhaps the ZINB model being slightly more accurate with its simulations of the frequency of the non-zero counts. Interestingly, however, there is a noticeable gap in the posterior value of p within the ZINB model compared to the ZIP model; here, we get a posterior value of $p = 0.623$, which suggests that the negative binomial distribution is able to explain more of the zero-inflation alone.

Whilst we have seen a significant difference in the performance between the two Poisson models, it is not immediately clear from this kind of analysis alone which of the competing models is the best fit for the data. This is, of course, is the power of the evidence, which will allow us to quantify the comparisons between models, and ultimately decide on which model fits the data best.

5.3 Model comparison: computing the evidence

Now that we have seen the different models applied to this example, we can look to compute the evidence for the different models. This is a large dataset with a non-trivial number of parameters, but as we have seen, we can still compute the evidence using the power posterior method with relative ease, using the same geometric temperature chain as we saw in Chapter 2. Table 5.1 shows the values of the log-evidence computed.

Model	Estimated log-evidence
$\mathcal{M}_1$	-8971.915
$\mathcal{M}_2$	-5900.029
$\mathcal{M}_3$	-5465.750
$\mathcal{M}_4$	-5447.424

Table 5.1: Values of the log-evidence estimates for the different models.

So far, we have calculated the evidence using a number of techniques, but of course we wish to use the evidence in order to perform direct model comparison. There are a number of ways in which a Bayesian might use the evidence in order to compare models generally, some of which may require models to be nested, for example the Savage-Dickey density ratio [Dickey, 1971], which looks to simplify Bayes factors by fixing parameters of more complex models in order to recover simpler, nested models. However, in order to ensure that our methods can be applied

to any model, we will look at some more general approaches to model comparison.

5.3.1 Bayes factors

Suppose that, in a more general situation, we wish to compare two different competing models, $\mathcal{M}_1$ and $\mathcal{M}_2$. Even before we see any data, we may have prior beliefs about which model is more likely to be the true data generating process, and we can quantify this through prior model odds [Gosling and Heaps, 2025]. We can define

$$O_f(\mathcal{M}_1) = \frac{\pi(\mathcal{M}_1)}{\pi(\mathcal{M}_2)} = \frac{\pi(\mathcal{M}_1)}{1 - \pi(\mathcal{M}_1)},$$

the odds in favour of $\mathcal{M}_1$. This of course is implied by the prior model probabilities $\pi(\mathcal{M}_1)$ and $\pi(\mathcal{M}_2)$. Once we have seen the data, we can then update our beliefs in a typical Bayesian way by combining the prior odds and the Bayes factor in favour of $\mathcal{M}_1$,

$$B_{12} = \frac{\pi(\mathbf{y}|\mathcal{M}_1)}{\pi(\mathbf{y}|\mathcal{M}_2)},$$

which clearly relies on the calculation of the evidence. If $B_{12} > 1$, then this means that the support for $\mathcal{M}_1$ given the data and prior specification is greater than that for $\mathcal{M}_2$, and conversely if $B_{12} < 1$. We can then calculate posterior odds in favour of $\mathcal{M}_1$:

$$O_f(\mathcal{M}_1|\mathbf{y}) = \frac{\pi(\mathcal{M}_1)}{\pi(\mathcal{M}_2)} \cdot B_{12}.$$

Indeed, we thus see that this is exactly what we would like: a way in which to quantitatively compare models in a Bayesian fashion.

For our data on the usage of grocery delivery services, we can iteratively compare subsets of two models at a time, working through possible pairs of models and eliminating one model at a time until we have a favoured model. Since we have log-evidence values, we can work on the log-scale here too and calculate the log Bayes factor:

$$\log B_{12} = \log \pi(\mathbf{y}|\mathcal{M}_1) - \log \pi(\mathbf{y}|\mathcal{M}_2).$$

Table 5.2 shows the values of the Bayes factors computed. We clearly see that, as expected, the Poisson model performs by far the worst, and then having removed it from contention, it is followed by the ZIP model before the ZINB model is shown the most support by the Bayes factor.

Model comparison	Log Bayes factor
$\log B_{12}$	−3071.886
$\log B_{13}$	−3506.165
$\log B_{14}$	−3524.491
$\log B_{23}$	−434.250
$\log B_{24}$	−452.576
$\log B_{34}$	−18.326

Table 5.2: Comparisons of various subsets of models using the log Bayes factor.

What is perhaps most striking is how decisive these values are; when we exponentiate again to recover the B_{ij} on the natural scale, we see that many of these values are negligible, which suggests that there is almost no support for most of these models at all in comparison to the ZINB model.

One particular comparison that is of interest is that between the ZIP model and the negative binomial model with no zero-inflation included. Whilst we already saw visually that the negative binomial model would perform

well, accounting for much of the zero-inflation naturally, the log Bayes factor of -434.25 gives almost no support to the ZIP model at all given the data. This suggests that the overdispersion in the data plays a more important role in which model will be more appropriate than the zero-inflation by itself; whilst the ZIP model was good at adapting to the inflated zeroes we saw in the data, the underlying Poisson distributional assumption works well in situations where the mean and variance of the data are well aligned, due to the natural behaviour of the Poisson distribution. That is, of course, not present for our data as we determined, and so the negative binomial model outperforms it here, since the incorporation of the dispersion parameter ϕ allows it to adjust for the difference in mean and variance.

When we then compare the negative binomial and ZINB models, we obtain a value for the log Bayes factor which is comparatively lower in magnitude, but even still it is extremely decisive, giving very little support to the non-zero-inflated model over its zero-inflated counterpart given the data. We could take this further by calculating the posterior model odds, however even a prior probability of close to 1 for $\mathcal{M}_3$ here would not lead the effect of the Bayes factor to be overturned. Ultimately, the data and likelihood have completely taken over any prior here, particularly given the volume of data observed.

5.3.2 Posterior model probabilities

Instead of comparing models in subsets of two, we could compare all of our competing models at once. Calculating posterior model probabilities directly, instead of odds, allows us to do this. Having defined prior model probabilities, we can define

$$\pi(\mathcal{M}_k|\mathbf{y}) = \frac{\pi(\mathcal{M}_k)\pi(\mathbf{y}|\mathcal{M}_k)}{\sum_{k=1}^m \pi(\mathcal{M}_k)\pi(\mathbf{y}|\mathcal{M}_k)},$$

again depending on the evidence. We will now need to specify a set of prior model probabilities across all of the competing models. Suppose that we had very little faith in the base Poisson model, but had little strong opinion on which of the remaining models might be correct. We might therefore set

$$\begin{aligned}\pi(\mathcal{M}_1) &= 0.1 \\ \pi(\mathcal{M}_2) &= \pi(\mathcal{M}_3) = \pi(\mathcal{M}_4) = 0.3\end{aligned}$$

so as not to favour one of the other models in particular. We will once again make use of the Log-Sum-Exp trick in order to evaluate these probabilities without running into numerical underflow issues since the log-evidence values are so small. The posterior model probabilities obtained are shown in Table 5.3.

Again, we see a similar result, in that the data have completely taken over the calculation, which is not unusual for a dataset of this size. This is still striking, however; this analysis suggests it is almost impossible that the true data were generated by a Poisson model, for example. What is perhaps most striking is how decisive this result is, but what is causing a result with such confidence?

Model	Posterior model probability
$\mathcal{M}_1$	0
$\mathcal{M}_2$	2.73×10^{-197}
$\mathcal{M}_3$	1.10×10^{-8}
$\mathcal{M}_4$	1

Table 5.3: Posterior model probabilities.

5.3.3 Not keeping an open mind: the $\mathcal{M}$ -closed scenario

What we see here is a result of an underlying assumption that we have made: that one of the four models is indeed the true model. This scenario, called the $\mathcal{M}$ -closed scenario, assumes that given a particular finite set of models $\mathcal{M}$, one of the models will be the true data generating process [Bernardo and Smith, 1994]. Whilst for this dataset, it appears that our ZINB model does indeed perform very well when synthetic data are generated from it compared to the true data, the reality of an $\mathcal{M}$ -closed world is that we are ruling out the possibility of any other models than the ones we present. Consequently, it is not surprising that the Bayes factor and posterior model probabilities lead us to prefer $\mathcal{M}_4$ in such a decisive way; it is the best fit of the four (to which a large sample size only assists the decision), and we have ruled out the possibility that any other model (of which there are infinitely many!) could be the true data generating process.

We could instead look to alternative perspectives, such as the $\mathcal{M}$ -complete scenario, in which we believe that a true data generating process does indeed exist, but that we cannot conceptualise it, or that the subset $\mathcal{M}$ of models is more computationally convenient, perhaps reducing the burden for further communication too. From this perspective though, specifying prior model probabilities $\pi(\mathcal{M}_k)$ does not make sense, which means that rather than attempting to select one of a finite number of competing models as the ‘truth,’ we might look to other methods, such as predictive model averaging, in order to incorporate knowledge from a range of different models. Ultimately, it is important to keep in mind that whilst our comparative analysis shows that model $\mathcal{M}_4$ is decisively preferred over the other models presented, it does not say anything about $\mathcal{M}_4$ compared to the wider ecosystem of possible models.

Now that we have seen a comparison within the $\mathcal{M}$ -closed world, we can look to analysis within the $\mathcal{M}$ -complete world.

5.3.4 Sanity check: a non-evidence-based approach

One metric we can use for analysing models in a quantitative way that avoids directly using the evidence is the Deviance Information Criterion (DIC). This criterion was first described by Spiegelhalter et al. [2002], and is one used to measure the predictive accuracy of a model as opposed to a direct comparison between whether one model represents the ‘truth’ or not. For a model with parameters θ and data $\mathbf{y}$, the definition is

$$\text{DIC} = \overline{D} + p_D,$$

where:

- $\overline{D}$ is the posterior mean deviance,

$$\mathbb{E}_{\theta|\mathbf{y}}(D(\theta)) = \mathbb{E}_{\theta|\mathbf{y}}(-2 \log \pi(\mathbf{y}|\theta))$$

- p_D is a measure of the number of effective parameters, the difference between the mean deviance and the deviance of the means,

$$p_D = \overline{D} - D(\bar{\theta}),$$

where $\bar{\theta}$ is simply an average of the posterior θ samples.

In a similar fashion as for other criteria, a lower value of the DIC indicates better predictive accuracy. For our data and models, the values of the DIC are presented in Table 5.4.

Model	DIC
$\mathcal{M}_1$	18092.529
$\mathcal{M}_2$	11885.799
$\mathcal{M}_3$	11012.449
$\mathcal{M}_4$	10969.150

Table 5.4: DIC values.

We see a similar pattern here too as we did for our evidence-based calculations; whilst the gap between the two overdispersed models is smaller, model $\mathcal{M}_4$ has the lowest DIC value, and thus better predictive ability using this metric.

DIC here has a slightly different objective to that we have seen from the Bayes factor and posterior model probabilities so far. In particular, this criterion is a measure of the predictive ability of the models, whereas our previous methods looked directly at

model comparison. This is a feature of the $\mathcal{M}$ -complete world, as Robert [2014] points out. Indeed, Robert is very much a critic of the DIC, arguing that the measure is still used as a direct comparison between models despite this, and that it also falls somewhat outside of the Bayesian paradigm. Instead, metrics such as the Widely Applicable Information Criterion (WAIC) [Watanabe, 2013] could be used, since these are fully Bayesian. Whilst the benefit of metrics like the WAIC or the evidence is that they are fully Bayesian, it is reassuring to see that the results in Table 5.4 are broadly in line with our expectations from the previous analysis.

5.3.5 Back to reality

Having performed all of this model analysis, what have we really learned from the evidence? Moving back towards the $\mathcal{M}$ -closed world, it certainly seems as though that among the four models presented for comparison, $\mathcal{M}_4$, our zero-inflated negative binomial model, performs the best and most closely matches the data. We can look to our original problem, that of which covariates had a significant impact on the response, the number of times American adults used grocery delivery services within a 30-day period. Firstly, let us consider the regression component of the model; that is, the counts that were not generated from the perfect state. Looking at the posterior estimates for the β_j in Table 5.5, we can assess the estimates themselves, and look at their significance by constructing 95% equi-tailed credible intervals. Of course, since we are dealing with a number of factor variables, we have kept the first level of each factor as the reference level. Note that the full encoding for each variable, with factor levels, is given in the Appendices.

From this analysis, we can credibly say that sex and an adult's employment status and medical condition from these variables have the greatest effect on the log expected counts, since the relevant 95% credible intervals do not contain zero. Women appeared to use these delivery services more often, with an expected increase of a factor of $e^{0.20} = 1.22$ times more counts compared to men, whilst those who work and those who faces no medical travel difficulties are less likely to use them, instead likely being able to visit brick and mortar stores more easily, perhaps on the commute to or from work.

Looking at the posterior $p = 0.623$, we see that a very large portion of the respondents would never use these services in the first place, perhaps not living within a region with access, or

β_j	Variable	Estimate	95% CrI
β_1	Intercept	1.36	[0.73, 2.04]
β_2	R_AGE	-0.01	[-0.01, 0.00]
β_3	R_SEX_2	0.20	[0.07, 0.34]
β_4	EDUC_2	-0.69	[-1.47, 0.05]
β_5	EDUC_3	-0.13	[-0.74, 0.46]
β_6	EDUC_4	0.06	[-0.55, 0.63]
β_7	EDUC_5	-0.07	[-0.68, 0.53]
β_8	EDUC_6	0.07	[-0.52, 0.64]
β_9	EDUC_7	0.07	[-0.54, 0.65]
β_{10}	EDUC_8	0.12	[-0.50, 0.73]
β_{11}	WORKER_2	-0.32	[-0.48, -0.17]
β_{12}	MEDCOND_2	-0.47	[-0.71, -0.25]
β_{13}	DRIVER_2	0.15	[-0.08, 0.40]

Table 5.5: Posterior negative binomial regression coefficient values.

without the means to take advantage of them (for example, not having access to a smartphone).

Now that we have seen some of the variables do not appear to affect the response much if at all, we could look to further model selection via backward selection if we wished, removing variables one-by-one to simplify further, whilst keeping the overall overdispersed and zero-inflated structure.

5.4 Summary

In this chapter, we have applied the evidence onto a real dataset looking at the effect of a number of covariates on the number of times grocery delivery services were used by American adults over a 30-day period. We applied the models introduced in Chapter 4 to this data, and introduced methods for direct model comparison that utilise the evidence. These methods generalise well to any set of models we might wish to compare, and they delivered extremely decisive results, even between the two negative binomial models, where it was more difficult to distinguish meaningful differences visually.

It is, however, important to consider the larger picture, and take into account the fact that approaching model comparison in this way at all and attempting to choose just one model to represent the ‘truth’ may be short-sighted: the $\mathcal{M}$ -closed world is convenient because it allows us to compare a small set of models very easily, but it is unlikely to reflect the real world. Indeed, we saw that the results of our model comparison were incredibly decisive, but this is ultimately the result of an arbitrary limit on the comparison; of course, given a model that better reflects the data, the Bayes factor and posterior model probabilities will inform us we should use that model compared to its inferiors. But that does not necessarily mean that the chosen model is correct. Indeed, whilst in this example Figure 5.5 suggests that our ZINB is a good reflection of the data, this dilemma shows that whilst the evidence is indeed incredibly powerful to compare models in the most general and widely-applicable way, we must not rely on it alone, or without context, lest we miss the bigger picture.

Chapter 6

Conclusion

In this dissertation, we have seen that computing the evidence is not a trivial problem, and introduced and used a number of different methods to calculate this critical quantity, assessing their applicability to the problem. We have also applied the evidence to both synthetic and real examples, looking at data which are complicated by their zero-inflation, and accounting for this.

Whilst we saw the power of the evidence in enabling us to directly compare competing models even for the most general cases where we lack conjugacy, and indeed we observed extremely decisive results, we also saw that the methods for this particular approach to model comparison could potentially be based in the idea that we are comparing a ‘complete’ set of models which account for all the models that could have generated the data. This is unlikely to be the case. This does not mean that the evidence is not without its merits (very far from it!), but that it serves as a reminder to contextualise our results: despite observing decisive model comparisons, we must assess whether we think our models could match the real world *at all*, not just in comparison to one another.

In terms of the computation of the evidence and its relation to model comparison, in Chapter 2 we saw a number of different methods to compute the evidence, including some naïve methods such as the Monte Carlo and harmonic mean estimators, which are simple to digest, but are undermined significantly by their lack of stability. We also introduced some more advanced techniques to calculate the marginal likelihood, Chib’s method and the power posterior method, whilst studying the results of applying these methods on a very simple set of examples to verify their validity.

Having verified their validity, in Chapter 3 we performed a systematic analysis on a Bayesian linear regression model in order to determine which of the methods was best: which gave the most accurate results, and which was the most easily and widely applicable to a general class of models. We saw that whilst the harmonic mean estimator was woeful, and the Monte Carlo estimator was relatively accurate for this particular model for a high number of samples, it was Chib’s method and the power posterior approach that were the most accurate and most reliable. Whilst Chib’s estimator was perhaps the most accurate, we also saw that it is significantly more involved to compute, particularly for high-dimensional models with the highest number of parameters as one is required to compute full conditional densities. Thus, we looked to the power posterior method in order to aid our subsequent analysis, since it is easy to implement and reliable given a sufficient number of samples.

To explore the power of the evidence in practice, in Chapter 4 we looked at a series of more complex setups to model zero-inflated count data, introducing models which can account for data which is generated by an “always-zero” or “perfect state” process. We introduced the relevant parametrisations for Poisson and negative binomial zero-inflation models, before showcasing the flexibility and versatility of the power posterior method in calculating the evidence.

Finally, in Chapter 5 we looked at a real dataset on American adults’ usage of grocery delivery

services, and applied our newfound models in order to assist with quantitative model comparison. We saw that we could use the evidence computed in order to calculate Bayes factors and posterior model probabilities, both of which decisively preferred the zero-inflated negative binomial model, which was able to account for the significant overdispersion within the data whilst still reflecting the zero-inflation well. The evidence empowered us to make strong decisions on the validity of models in comparison to each other but, as aforementioned, we must also consider the models' validity within the wider set of possible models independently.

Whilst we have explored a number of different methods and applications for computing the evidence, this is by no means an exhaustive overview. For further work, we could look to other methods for computation, such as Annealed Importance Sampling or Sequential Monte Carlo, which are other widely-used methods in the literature and can also be used to efficiently and accurately compute the marginal likelihood. We could also look to move to an $\mathcal{M}$ -complete world, where it is not necessarily assumed that one of the models presented will be the true data generating process; in doing this, we could evaluate differing methods for combining information from an array of different models, such as predictive model averaging, which would help us in developing the most accurate models possible, and in justification to experts and the wider public.

Appendices

A.1 Grocery data variable encoding

The data used throughout Chapter 5 for analysis contains 6 explanatory variables, of which 5 are factor variables. The levels of the factor variables, along with their encoding [U.S. Department of Transportation, 2023], are given below:

- R_SEX: Respondent sex
 - R_SEX_1: Male
 - R_SEX_2: Female
- EDUC: Respondent’s highest level of education
 - EDUC_1: Less than high school
 - EDUC_2: Some high school, no diploma
 - EDUC_3: High school graduate, have diploma
 - EDUC_4: Some college, no degree/some trade school
 - EDUC_5: Associates degree/trade school certificate
 - EDUC_6: Bachelor’s degree
 - EDUC_7: Master’s degree
 - EDUC_8: Professional/Doctorate degree
- WORKER: Respondent’s employment status
 - WORKER_1: Worker
 - WORKER_2: Not worker
- MEDCOND: Condition or disability that makes travel difficult
 - MEDCOND_1: Yes
 - MEDCOND_2: No
- DRIVER: Respondent’s driver status
 - DRIVER_1: Yes
 - DRIVER_2: No

A.2 Code repository

The GitHub code repository for this dissertation can be found at

<https://github.com/alexeveresthouse/evidence>

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